

MSc Dissertation Report

" HR Analytics: Predicting and Addressing Employee Attrition with Machine Learning"

A dissertation submitted in partial fulfilment of the requirements of Sheffield Hallam University for the degree of Master of Science in Artificial Intelligence

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ABSTRACT

Employee attrition is a significant problem for businesses in today's cutthroat business climate, affecting financial resources, morale, and productivity. It is now more critical than ever for companies to be able to anticipate and handle employee turnover, primarily as they work to keep top talent and ensure business continuity. The application of machine learning (ML) techniques to forecast employee attrition and pinpoint important variables influencing voluntary resignations is examined in this research. This study offers data-driven insights and practical tactics for enhancing employee retention by utilizing HR analytics. With attrition as the target variable, the study uses a comprehensive HR dataset that includes 37 factors, such as demographic information, work-life balance scores, job satisfaction indicators, remuneration, and employment history. To guarantee data quality and robustness, preprocessing methods such as feature engineering, addressing missing values, and encoding categorical variables are used. To assess how well they predict employee attrition, several supervised machine learning techniques are used, including Random Forest (RF), Decision Tree (DT), Support Vector Machines (SVM), Gradient Boosting (GB), Logistic Regression (LR), and Artificial Neural Networks (ANN). Metrics including accuracy, precision, recall, F1-score, and area under the curve (AUC) are used to evaluate the models.

With Random Forest and Gradient Boosting emerging as top performers, attaining over 85% accuracy across validation tests, the results show that machine learning models can effectively forecast employee attrition. Key attrition factors, such as years since the last promotion, work-life balance, job satisfaction, and monthly pay, are revealed via feature importance analysis. Workers who experience stalled professional advancement, poor work-life balance, and poorer job satisfaction are likelier to quit the company. The study emphasizes how vital HR analytics are in helping businesses spot individuals who are at risk and take proactive measures to solve their issues.

Additionally, this dissertation adds to the body of literature by addressing employee attrition. This topic has historically been studied through qualitative evaluations by fusing machine learning techniques with HR statistics. In contrast to traditional methods, this study uses a quantitative, data-driven methodology that yields unbiased, valuable conclusions. Organisations can use the results to improve workforce stability, increase employee engagement, and prioritise retention tactics.

The study also examines the ethical issues surrounding the use of employee data to ensure privacy, confidentiality, and adherence to data protection laws. It gives HR managers advice on how to apply

predictive models in practical contexts, focusing on integrating machine learning tools with current HR management systems to dynamically monitor employee satisfaction and retention risk.

This study highlights the revolutionary potential of machine learning in HR analytics for forecasting staff turnover and creating successful retention plans. By utilising data-driven insights, businesses may reduce attrition, create a happy workplace, and succeed in the long run. To further improve prediction accuracy, future research can investigate including more behavioural and external economic components.

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List of Abbreviations

- ML – Machine Learning
- HR – Human Resources
- IT – Information Technology
- AI – Artificial Intelligence
- LLMs – Large Language Models
- LMS – Learning Management Systems
- RUL – Remaining Useful Life
- EDA - Exploratory Data Analysis
- ROI - Return On Investment
- HRM - Human Resource Management
- TAM - Technology Acceptance Model
- RBV - Resource-Based View
- NPL – Natural Language Processing
- IBM – International Business Machines Corporation
- RNNs – Recurrent Neural Networks
- CNNs - Convolutional Neural Networks
- XAI – Explainable Artificial Intelligence
- K-NN - K-Nearest Neighbours
- SVM - Support Vector Machines
- GDPR - General Data Protection Regulation
- ADA Boosting – Adaptive Boosting
- EDA - Exploratory Data Analysis
- CRISP-DM - Cross-Industry Standard Process for Data Mining
- R&D – Research and Development
- NumPy – Numerical Python
- CSV – Comma-Separated Values
- SMOTE - Synthetic Minority Oversampling Technique
- PCA – Principal Component Analysis
- PII – Personally Identifiable Information
- RBF – Radial Basis Functio

CHAPTER 1: INTRODUCTION

Introduction

Organisations in all sectors face a serious problem with employee attrition, which has a big impact on staff morale, financial stability, and operational effectiveness. High attrition rates could harm organisational knowledge and culture, interfere with processes, and raise the expense of hiring and training new employees. To forecast and handle attrition, HR practitioners have historically depended on statistical techniques; however, these methods frequently fall short of capturing the complex and dynamic nature of contemporary workforces. Machine learning (ML) has become a game-changing technique in recent years, providing precise and scalable answers to attrition problems.

The potential of machine learning to more accurately forecast attrition through the analysis of complex and varied data has been shown by research in this field. (Shete & Patil, 2021), for example, created machine learning algorithms that can precisely monitor attrition patterns in various industries, highlighting the significance of differentiating between true positives and false positives. Similarly, (Joseph et al., 2021) highlighted the importance of psychological aspects in turnover by integrating depression analysis and emotional assessments into attrition prediction models. The effectiveness of several machine learning techniques, such as random forests, logistic regression, decision trees, and artificial neural networks, has been assessed through comparative studies, such as those conducted by (Rustam et al., 2021) and (Mansor et al., 2021), which have highlighted the advantages and disadvantages of each strategy.

In addition to comparing models, other research has customised machine learning solutions to meet the needs of certain organisations or industries. (Sadana & Munnuru, 2021), for instance, examined attrition in IT companies and stressed the significance of early employee dissatisfaction identification to allow for prompt solutions. Similarly, (Habous et al., 2021) used supervised learning models to show how both personal and professional characteristics affect attrition rates. Together, these studies show how feature engineering and data preparation are becoming increasingly important for making correct predictions.

In order to increase employee engagement and enhance decision-making, recent studies have looked into integrating artificial intelligence (AI) with HR procedures in addition to attrition prediction. (Ma et al., 2024) emphasised the potential of big language models to analyse communication patterns and detect subtle signs of turnover intentions, (Okatta et al., 2024) stressed the relevance of AI and ML in personalising employee experiences. These developments mark a paradigm change in HR

analytics, indicating a move away from reactive to proactive methods of labour dynamics management.

By investigating the predictive potential of machine learning in addressing employee attrition, this dissertation aims to add to this growing corpus of knowledge. Utilising modern analytical tools, the study seeks to identify the primary causes of employee attrition, create precise prediction models, and provide workable retention solutions. The knowledge acquired will not only improve organisational effectiveness but also lay the groundwork for further HR analytics study.

Background of HR Analytics and Its Significance

A revolutionary change in how businesses handle workforce management is represented by HR analytics, often known as human resources analytics. In the past, HR procedures mainly depended on experience-based decision-making and intuition. However, HR analytics has become a vital tool for coordinating workforce strategies with organisational objectives since the introduction of data-driven techniques. HR specialists can find practical insights that assist the company and its employees by utilising data and sophisticated analytical methods. One of the main areas where HR analytics has shown great promise is in tackling employee churn, a widespread issue with broad ramifications.

Employee turnover, also known as employee attrition, is the rate at which workers depart from a company. Excessive attrition rates can have serious operational and financial consequences, such as higher training and recruitment expenses, hampered productivity, lower morale, and the loss of institutional knowledge. Since these issues call for proactive methods to anticipate and reduce turnover, HR analytics are a vital tool for contemporary businesses.

The use of machine learning techniques, which provide unmatched predictive modelling capabilities, has been a hallmark of the advancement of HR analytics. Traditional statistical techniques frequently fail to capture the intricate interactions between work-life balance, career growth possibilities, and job satisfaction that affect turnover. Machine learning, on the other hand, is excellent at analysing big and varied datasets, finding non-linear patterns, and making precise predictions. For instance, (Krishna & Sidharth, 2022) created a machine learning model that can precisely forecast employee turnover rates using HR analytics data from Kaggle. In a similar vein, (Jayakumar et al., 2023) emphasised the significance of machine learning algorithms and data pretreatment methods in forecasting attrition according to job titles.

Deeper insights into employee behaviour have been made possible by machine learning approaches, which go beyond prediction accuracy. A cluster-based study on HR analytics used machine learning to analyse employee turnover dynamics, highlighting the importance of data processing in producing

useful insights. The use of large language models (LLMs) is one recent development that has significantly broadened the scope of HR analytics. To illustrate the usefulness of these tools in tackling workforce concerns, research have demonstrated how LLMs may analyse employee communication patterns to discover subtle indications indicative of turnover intentions.

According to (*The State of HR Analytics Today (2024)* | by Eric Sandosham, Ph.D. | Medium, n.d.), agile HR analytics emphasises how crucial it is for HR management to embrace adaptable and data-driven strategies. Employers can identify at-risk workers, customise retention plans, and eventually raise employee satisfaction by utilising machine learning and predictive modelling. This is consistent with other studies that highlight the importance of data-driven decision-making in enhancing organisational performance and retention.

Although there is no denying HR analytics' potential, there are obstacles to its uptake. Significant obstacles are presented by problems including data privacy, integration between various technologies, and organisational resistance to change. But these obstacles are still being addressed by the expanding corpus of research and technological developments, which makes HR analytics a vital component of strategic labour management.

To sum up, HR analytics is a paradigm change in HR management, and one of its most notable applications is the capacity to anticipate and control employee attrition. Organisations can transition from reactive to proactive methods and build a more stable and engaged workforce by utilising machine learning. Organisations hoping to stay competitive in a constantly shifting business environment will need to embrace these developments as the area develops.

Importance of Understanding Employee Attrition in Organizations

Organisations are concerned about employee attrition because it affects staff retention, operational effectiveness, and financial performance. Reduced team cohesion, institutional knowledge loss, and recruiting and training expenses are all consequences of high turnover rates. Maintaining worker stability and guaranteeing organisational success need an understanding of attrition.

Machine learning has become a potent instrument for attrition prediction and management, providing higher scalability and accuracy than conventional techniques. Numerous models, such as Decision Trees, Random Forests, and Gradient Boosting, have been shown to be successful in forecasting employee turnover in studies like (Shete & Patil, 2021) and (Qutub et al., 2021). In a similar vein, (Sadana & Munnuru, 2021) and (Yahia et al., 2021) stressed the significance of using data to pinpoint

elements that affect attrition, such as work-life balance and job satisfaction. These insights enable HR professionals to devise targeted retention strategies.

The capacity of machine learning models to rank important predictions has improved decision-making even further. For example, (Raza et al., 2022) created an optimised Extra Trees Classifier for accurate turnover predictions, while (Mitravinda & Shetty, 2022) used HR analytics data to evaluate the significance of features causing attrition. These developments highlight how data-driven strategies might help reduce attrition issues.

Beyond the technical viewpoint, comprehending attrition is consistent with more general organisational objectives. In addition to keeping important personnel, companies that deal with turnover also enhance their employer brand and create a positive work atmosphere. Businesses may preserve institutional knowledge, increase employee satisfaction, and boost long-term performance by proactively controlling attrition.

The work taken as a whole, emphasises the need of comprehending employee attrition and the revolutionary possibilities of machine learning. Organisations can achieve workforce stability and lower turnover by identifying critical elements and putting data-driven plans into practice. This will lead to success over the long term.

Role of Machine Learning in Predicting and Addressing Attrition

In recent years, the use of machine learning to anticipate and manage attrition has become a game-changing strategy. Machine learning models have proven to be effective in workforce management by analysing intricate data patterns and offering practical insights to reduce staff turnover.

Machine learning's potential goes beyond employee turnover; creative implementations in a variety of industries offer insightful lessons for HR analytics. (Fahd et al., 2021) investigated how identifying at-risk students through the analysis of student interaction data using Learning Management Systems (LMS) allows for prompt interventions to improve academic performance. This is similar to the proactive retention tactics used by businesses to deal with disgruntled workers. The use of machine learning to predict attrition in longitudinal research was also confirmed by (Jankowsky & Schroeders, 2021), who emphasised the significance of model dependability and data generalizability—two crucial elements in employee attrition prediction.

The use of machine learning in other fields, like engineering, healthcare, and education, further demonstrates its adaptability. For instance, (Sun et al., 2022) used transfer learning to forecast the Remaining Useful Life (RUL) of rolling bearings, demonstrating the efficacy of predictive

maintenance techniques, while (Kwofie et al., 2023) illustrated its role in addressing high attrition rates in drug discovery pipelines. By utilising data-driven insights, these approaches highlight the ability of machine learning to handle attrition issues in many scenarios.

These developments encourage HR practitioners to use machine learning methods for predictive analytics in the workforce sector. Organisations can gain a deeper understanding of the root reasons of attrition by combining personnel data, including performance indicators, job satisfaction surveys, and turnover trends. Additionally, novel techniques like probabilistic modelling and transfer learning may improve prediction accuracy, particularly when working with sparse or noisy data.

All things considered, the research shows how machine learning may revolutionise attrition issues, ranging from school performance and drug development to staff turnover. Machine learning helps firms retain important personnel, increase staff stability, and boost overall performance by facilitating proactive decision-making and predictive analytics. These observations emphasise how crucial it is to use cutting-edge analytical methods in order to handle the intricacies of attrition in the ever-changing organisational environment of today.

Research Question

What are the main causes of employee churn in the company, and how can predictive analytics be used to pinpoint vulnerable workers and suggest successful retention tactics?

Research Objectives

The main objective of this dissertation is to utilize data-centered methods to examine and forecast employee turnover, with the final aim of offering practical insights to improve workforce retention. This goal is backed by the subsequent objectives:

- To analyse trends and patterns in employee turnover and determine the key risk factors affecting attrition.
- To create machine learning models that can effectively forecast employee turnover by utilizing past workforce data.
- To pinpoint essential elements leading to attrition, including job satisfaction, travel for work, and frequency of promotions, and evaluate their significance.
- To offer strategic suggestions for decreasing turnover and enhancing employee engagement, retention, and overall organizational effectiveness.

These goals correspond with the larger aim of using HR Analytics to facilitate decision-making based on data in human resource management.

Deliverables

This dissertation will provide a thorough exploratory data analysis (EDA) focusing on factors such as age, work-life balance, and job satisfaction, along with predictive models like Random Forest to identify at-risk employees. It will include an analysis of key attrition-influencing features with visual aids and a recommendations report offering actionable strategies to address significant drivers of attrition, such as managing overtime and optimizing promotion schedules.

CHAPTER 2: LITERATURE REVIEW

Overview of employee attrition and retention strategies

Employee attrition and retention are major issues for organisations worldwide, and they have a significant effect on worker morale, organisational effectiveness, and financial stability. Attrition, which can be separated into two categories—voluntary and involuntary turnover—is the pace at which employees leave an organisation. Because it exposes underlying problems within the company, voluntary attrition—when workers decide to quit for reasons like discontent or better opportunities—presents a bigger challenge for HR departments ('A Review of Employee Retention Strategies: Implication for Future Research', 2021).

In order to lessen the detrimental effects of attrition, effective retention techniques are necessary. According to research, it's critical to address both personal and professional aspects that affect an employee's decision to stay or quit. Fostering job happiness, enhancing organisational commitment, and establishing favourable work environments are important strategies ('A Review of Employee Retention Strategies: Implication for Future Research', 2021). Organisations must prioritise comprehensive strategies that include professional development, work-life balance, and a supportive work environment in addition to competitive remuneration and financial perks (Enamala & Reddy, 2022). According to (Sridevi & Reddy, 2023), these components support a strong employee value proposition, which has been demonstrated to be an effective retention strategy.

In sectors like IT, where fierce rivalry for talent, quick technological breakthroughs, and the dynamic nature of work are the main causes of high turnover rates, retention issues are more noticeable (Sridevi & Reddy, 2023). HR departments must create and execute creative policies that are suited to the particular requirements of their employees in order to handle this. The necessity for individualised retention tactics is highlighted by research showing that factors including age, career stage, and personal goals have a substantial impact on employees' intentions to stay (Sridevi & Reddy, 2023); (Karumuri, 2014).

Theoretical frameworks like Maslow's Hierarchy of Needs and Herzberg's Two-Factor Theory offer important insights into employee motivation by highlighting the importance of intrinsic elements like meaningful work, growth opportunities, and acknowledgement in lowering turnover. Furthermore, new developments in predictive modelling and HR analytics give businesses strong instruments to assess the success of retention plans and spot attrition threats. Organisations can take proactive steps to improve employee commitment and happiness by utilising data-driven insights.

Ultimately, keeping a steady and motivated workforce requires both well-thought-out retention measures and a thorough awareness of the issues causing attrition. Organisations must maintain flexibility in modifying their strategies for staff retention as industries change, guaranteeing that they are in line with the evolving demands and expectations of their workforce.

Significance of Predicting Employee Attrition

Organisations' current retention strategies use a diversified strategy to lower turnover and preserve a competitive edge. Effective retention requires addressing both extrinsic and inner motivators, as companies are aware. Career development opportunities and fostering a happy work environment have become goals, even though competitive compensation and benefits are still vital (Malhotra, 2018; Suriati et al., 2024a)(Cloutier et al., 2015; Haque, 2023). These strategies are predicated on the notions of employee pleasure and engagement, which are critical for reducing turnover and enhancing organisational efficacy.

The implementation of flexible work hours and wellness initiatives by firms is evidence that work-life balance is a crucial element of modern retention strategies (Malhotra, 2018).

In order to promote a culture of gratitude and openness, which raises job satisfaction, employee recognition initiatives and good communication are also essential (Cloutier et al., 2015). Research highlights the need of developing an inclusive corporate culture that fosters cooperation and creativity while employing a varied workforce with the necessary capabilities (Haque, 2023).

Employee commitment and engagement have been found to be significantly influenced by leadership approaches, especially transformational leadership (Suriati et al., 2024a). By establishing a feeling of purpose and connecting personal aspirations with corporate goals, transformational leaders motivate their workforce. Furthermore, ongoing employee engagement initiatives and feedback systems foster a sense of belonging and worth among staff members, which enhances retention (Malhotra, 2018); (Haque, 2023).

One important advancement in recent years has been the incorporation of technology into retention methods. The use of HR analytics and machine learning techniques to track employee sentiment, spot trends in disengagement, and tailor retention campaigns is growing. With the use of these tools, businesses may customise career paths and perks to meet the needs of each individual employee by making data-driven decisions. Organisations are using innovation to improve the employee experience, as demonstrated by AI-driven career planning tools and flexible benefits packages (Suriati et al., 2024b).

These tactics are effective, yet there are still issues. Implementing thorough retention initiatives may be limited by financial limitations, changing employee expectations, and cultural resistance. Furthermore, industry-specific considerations frequently call for specific approaches. For instance, although the IT sector concentrates on skill development to handle swift technological developments, the healthcare sector might place a higher priority on mental health support.

In summary, contemporary retention tactics demonstrate a comprehensive strategy that combines modern technologies with time-honoured methods. By emphasising employee growth, engagement, and pleasure, companies hope to build a steady and effective staff. The use of advanced analytics and inclusive leadership techniques will probably become more and more important in improving organisational success and employee well-being as retention strategies continue to change.

Role of HR Analytics in Workforce Management

HR analytics has become a vital tool for efficient personnel management, allowing businesses to make data-driven choices that improve strategic results and operational effectiveness. HR analytics gives HR practitioners the ability to forecast important workforce indicators, analyse employee data, and calculate the return on investment (ROI) of strategic HR efforts by offering statistical tools and predictive models (Ragimol et al., 2021). Leaders may better match workforce plans with organisational goals by using workforce analytics, which guarantees that the correct people is accessible when needed to satisfy company demands (Huselid, 2018).

As a subset of HR analytics, predictive analytics enables businesses to maximise employee retention, satisfaction, and performance. Particularly in positions involving direct interaction with customers, these measures are strongly associated with customer satisfaction (Alabi et al., 2024). For example, HR teams can take preventive measures by using workforce analytics to spot trends in employee behaviour, such as engagement levels or attrition risks. HR practitioners may develop an engaged and productive workforce by coordinating employee performance with overarching corporate objectives (Alabi et al., 2024).

HR analytics are used in many areas, such as hiring, where data-driven insights are used to find applicants who have the best chance of succeeding and sticking with the company. Analytics helps identify skill shortages in workforce planning, allowing for focused training and development programs. Additionally, by revealing possible biases in hiring or promotion procedures and offering metrics on demographic representation, HR analytics aids diversity and inclusion initiatives.

The application of HR analytics is not without difficulties, notwithstanding its benefits. Significant obstacles include the need for specialised skills within HR teams, system integration challenges, and

data protection concerns (Alabi et al., 2024). To ensure the ethical use of employee data while upholding trust and legal compliance, organisations must carefully negotiate these obstacles.

Artificial intelligence (AI) and machine learning are two examples of emerging technologies that are positioned to transform HR analytics by providing more sophisticated labour management capabilities. Large datasets can now be analysed in real time thanks to these technologies, which also reveal hidden patterns and produce actionable insights with previously unheard-of accuracy. AI-powered solutions, for instance, can use natural language processing to assess employee attitude or historical trends to forecast future workforce requirements. By encouraging a more flexible and responsive corporate culture, these developments promise to not only enhance labour management but also alter customer service standards (Alabi et al., 2024).

In summary, HR analytics provides strong tools to match organisational objectives with human resources, making it a fundamental component of contemporary workforce management. Organisations may maximise employee performance, improve retention, and attain long-term success by utilising predictive analytics and emerging technology. To fully realise the potential of HR analytics in the changing labour management landscape, however, issues with data protection and talent gaps must be resolved.

Machine Learning Applications in HR

Machine learning (ML) applications in human resource management (HRM) are revolutionising traditional approaches by employing data-driven and automated tactics for a variety of HR functions. From talent development and employee engagement to hiring and performance management, ML-driven solutions are assisting companies in making smarter, more efficient decisions in a number of areas (Garg et al., 2021; Rathore & Rathore, 2024a). These developments are supported by theoretical frameworks like the Technology Acceptance Model (TAM), which describes how organisations accept innovative methods like machine learning, and the Resource-Based View (RBV), which stresses using technology to gain strategic benefits.

By automating resume screening, lowering bias, and increasing diversity in hiring, machine learning algorithms are revolutionising the candidate selection process in the recruitment industry (S, 2023). In order to guarantee that companies have the best personnel in key positions, predictive analytics helps identify high-potential individuals and facilitates efficient succession planning (S, 2023). Additionally, ML overcomes the drawbacks of conventional, standardised approaches by facilitating the creation of learning programs and professional progression pathways that are customised to each employee's needs (Rathore & Rathore, 2024b).

Sentiment analysis and natural language processing (NLP) are improving employee feedback systems, allowing companies to measure employee sentiment and promote open communication (Rathore & Rathore, 2024b). To find regions of discontent or possible attrition concerns, for instance, ML-powered sentiment analysis systems can examine staff surveys and open-text responses. Proactive steps to improve employee satisfaction and engagement are informed by these insights. ML applications in HR go beyond workforce strategy to operational effectiveness. Real-world case studies from businesses like Hilton, Xerox, and IBM offer quantifiable benefits including higher productivity, lower attrition rates, and enhanced employee engagement (Saxena et al., 2023). For instance, Hilton improved the candidate experience and accelerated the hiring process by using machine learning algorithms.

In a similar vein, IBM greatly decreased turnover by using predictive analytics to identify employees who were at danger of leaving and putting targeted retention initiatives into place.

Although there are clear advantages of integrating ML into HRM, such as increased decision-making, cost savings, and objectivity, there are drawbacks as well. Significant obstacles still exist because of ethical issues such as algorithmic bias, data privacy, and transparency (S, 2023; Saxena et al., 2023). Maintaining employee trust and legal compliance requires that ML algorithms be developed and applied equitably, with strong data governance mechanisms.

The use of generative AI to develop individualised employee experiences and tools for managing remote and hybrid workforces are examples of emerging developments in machine learning for human resource management. These developments are in line with the global trend towards workplaces that are more adaptable and employee-focused. Furthermore, industry-specific adoption trends show differences in ML applications, with industries like IT and healthcare setting the standard because of their reliance on highly qualified and flexible workers.

In summary, by improving productivity, precision, and customisation across essential tasks, machine learning is transforming human resource management. To fully realise ML's promise while preserving employee confidence, companies must address ethical issues and promote transparency as they continue to investigate its uses. Organisations may create more engaged and resilient workforces and secure long-term success in a business environment that is changing quickly by combining machine learning (ML) with conventional HR procedures.

Review of Existing Predictive Models for Attrition

The increasing use of machine learning methods to tackle workforce stability issues is highlighted by recent studies on employee attrition prediction models. These models provide sophisticated features for examining sizable and intricate datasets, allowing businesses to spot trends and more precisely forecast turnover concerns. According to comparative research, algorithms like Random Forest and XGBoost routinely perform better than alternative methods because of their capacity to manage big data sets and non-linear connections (Alqahtani et al., 2024). Even while these models perform quite well in terms of prediction, there may be difficulties with their interpretability, especially for HR professionals looking for practical insights.

The IBM HR Analytics dataset, which offers a consistent framework for performance benchmarking, is still a popular option for model creation and testing. Despite its extensive use, issues with representativeness and data reliability still exist since real-world datasets frequently contain missing values and noise, which makes prediction attempts more difficult (Al-Alawi & Ghanem, 2024). To guarantee solid and trustworthy outcomes, researchers stress the value of using a variety of models and assessment metrics, including precision, recall, and F1-score (Al-Alawi & Ghanem, 2024).

The potential of emerging methods to improve interpretability and forecast accuracy is making them popular, such as deep learning models and CatBoost (Kambhampati & Rao, 2024a). CatBoost is especially applicable to HR analytics since it was created to manage categorical data effectively and has proven to perform better in situations involving mixed data types. Although their transparency and computing requirements continue to be major obstacles, deep learning models like recurrent neural networks (RNNs) and convolutional neural networks (CNNs) show promise in capturing complex interactions and temporal patterns.

The incorporation of many components into analytical frameworks is a significant trend in attrition prediction research. In order to increase the comprehensiveness and relevance of the model, studies emphasise the importance of including employee demographics, job characteristics, organisational culture, and even external economic factors (Kambhampati & Rao, 2024a). This all-encompassing strategy is in line with HR analytics' growing significance as a tactical instrument for workforce management.

In attrition prediction research, ethical issues and data privacy are becoming more and more important. Consent, bias, and fairness are issues that arise when using employee data, especially when algorithms are used to make decisions that impact people's careers (Banerjee, n.d.; Kambhampati & Rao, 2024a). Strong data governance procedures, openness in model design, and commitment to

moral principles are necessary to address these problems in order to preserve employee confidence and legal compliance.

Despite the tremendous progress the sector has achieved, there is still room for more innovation. In order to improve model transparency and foster confidence among HR experts, future research could concentrate on incorporating explainable AI (XAI) methodologies. The restrictions of dataset specificity may also be addressed by applying transfer learning, which could allow models to benefit from insights from one organisational environment to another.

In summary, the analysis of current attrition prediction models demonstrates the quick development of machine learning methods and their revolutionary potential for HR analytics. Researchers and practitioners can further improve these tools to enable proactive workforce management and strategic decision-making by tackling issues with data quality, model interpretability, and ethical considerations.

Identification of Gaps in Literature

Significant advancements have been made recently in the use of classification algorithms including logistic regression, decision trees, and random forests in HR analytics research aimed at forecasting employee turnover through machine learning. These techniques have achieved remarkable predicted accuracy rates while identifying important attrition-causing elements. For example, it has been reported that random forest classifiers can reach up to 98% accuracy (Reshma et al., 2020). To improve both predicted accuracy and model interpretability, cutting-edge methods like CatBoost and deep learning models are also being investigated (Kambhampati & Rao, 2024b). Furthermore, research has indicated that in order to develop more thorough attrition models, it is critical to use a variety of data sources, including organisational aspects, job characteristics, and employee demographics (Kambhampati & Rao, 2024b).

Even with these developments, there are still a number of gaps in the literature. The extensive reliance on publicly accessible datasets, like the IBM HR Analytics dataset, which might not accurately reflect the complexity and diversity of actual organisational contexts, is one major drawback. These datasets frequently lack the depth of data required to capture subtle elements that can affect employee attrition, such as industry-specific dynamics, economic situations, and cultural influences.

The restricted generalisability of current models is another gap. It is difficult to apply many models across industries or geographical locations since they are customised for particular datasets or organisational contexts, even when studies have shown excellent accuracy rates. Additionally, even though innovative techniques like CatBoost and deep learning have demonstrated potential, nothing

is known about how to actually use them in HR situations. Significant obstacles are presented by problems like these intricate models' interpretability and the processing power needed to implement them.

Although they are becoming more widely recognised, ethical issues and data protection are not fully covered in the literature. For example, there isn't much talk about preventing algorithmic bias or making sure that predictive models' influence on HR choices is transparent. These concerns are especially important when it comes to employee attrition because forecasts have a direct effect on people's careers and company policy.

Furthermore, there is also a lack of research on the integration of external factors including competitive market conditions, macroeconomic trends, and employee attitude analysis. By adding such data, attrition models may become more applicable and predictive. Similarly, few studies systematically assess the trade-offs between accuracy, interpretability, and practical usability in real-world HR scenarios, despite the fact that researchers support the use of many evaluation metrics and machine learning models.

In summary, even if recent developments in machine learning have greatly improved HR analytics' capacity for prediction, there are still glaring gaps in the literature that demand more research. To create reliable and useful attrition prediction models that satisfy the requirements of various organisations, it will be essential to address these constraints, which include enhancing model generalisability, incorporating cutting-edge methodologies, and addressing ethical issues.

Rationale for the Study

Organisations face a serious problem with employee attrition, which is the gradual loss of valuable personnel (S. U, 2020; D. B. Lavanya, 2017). In addition to upsetting organisational stability, high attrition rates result in severe financial expenses, such as those related to hiring, selecting, and training replacements (S. U, 2020). A major area of attention for human resource management, attrition affects institutional knowledge, staff morale, and overall productivity in addition to the financial consequences (D. B. Lavanya, 2017). Maintaining organisational performance and competitive advantage requires good management of attrition, even though some degree of it is unavoidable.

Conventional methods of researching attrition have depended on statistical methods such multiple regression, correlation analysis, and factor analysis (D. B. Lavanya, 2017). These techniques have yielded important information about the behaviours of job seekers and how they relate to attrition rates. However, traditional models are limited in their ability to capture the intricate interactions between many factors impacting attrition due to the dynamic and multidimensional structure of

modern workforces. For instance, it is challenging to predict employee demographics, job satisfaction, work-life balance, and organisational culture just using linear methodologies.

Machine learning developments present revolutionary possibilities for resolving these issues. Large, complicated datasets can be processed using machine learning models, which can also reveal hidden patterns that conventional approaches might miss. Through the integration of many data sources, including job characteristics, personnel demographics, and performance indicators, machine learning helps organisations to better predict attrition and create focused retention strategies. In today's data-rich environment, where businesses are using predictive analytics more and more to make proactive decisions, these skills are especially important.

This study is justified by its ability to close the gap between new technological advancements and conventional HR procedures. Even while previous studies have shown that machine learning models are effective at predicting attrition, issues like model interpretability, ethical issues, and data privacy are still not well understood. By creating a thorough framework for utilising machine learning to anticipate and manage staff attrition while taking into account the moral and practical ramifications of its use, this study seeks to address these problems.

Additionally, this study addresses the increasing need for workforce management insights that can be put into practice. The study helps to lower turnover, increase employee engagement, and promote long-term organisational success by giving businesses innovative methods to detect at-risk workers and carry out prompt interventions. By investigating cutting-edge uses of machine learning in HR analytics, it also adds to the body of knowledge in academia and opens the door for more studies in this area.

To sum up, this study is relevant and current since it uses cutting-edge technology to solve a significant organisational problem. The goal is to offer a strong foundation for comprehending and controlling employee turnover in a business environment that is changing quickly by fusing the strategic insights of HR analytics with the predictive capability of machine learning.

CHAPTER 3: RESEARCH METHODOLOGY

Research Design, Philosophy, and Approach

Research design, philosophy, and approach form the foundational framework of any academic inquiry, guiding the selection of appropriate methodologies and ensuring the research aligns with its objectives. This chapter outlines the philosophical underpinnings, methodological choices, and systematic approaches employed in this dissertation to address the research question: “How can predictive analytics be used to identify at-risk individuals and suggest successful retention strategies?” The primary goal is to predict employee attrition using machine learning techniques while deriving actionable insights to enhance retention strategies. By adopting a well-defined research design, this study ensures reliability, validity, and relevance in its findings.

Research Design

The research design adopted in this study is explanatory and quantitative in nature. Explanatory research is appropriate given the aim to explore relationships between variables influencing employee attrition, identify key risk factors, and develop predictive models. A quantitative approach aligns with the data-driven nature of machine learning, emphasizing measurable evidence and statistical analysis. The study relies on a publicly available dataset, “Capstone Project: IBM Employee Attrition Prediction,” sourced from Kaggle. The dataset comprises 37 features covering demographic, organizational, and performance-related variables. This structured dataset facilitates comprehensive data preprocessing, feature engineering, and subsequent model development. Machine learning techniques such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (K-NN), Support Vector Machines (SVM), Kernel SVM, Naive Bayes, Gradient Boosting, and ADA Boosting are employed to predict employee attrition. Model performance is evaluated using standard metrics, including confusion matrices, to ensure robust and replicable results. Ethical principles guide the use of data, focusing on privacy and compliance with regulations such as the General Data Protection Regulation (GDPR). Although the dataset is simulated, ethical considerations are addressed to reflect real-world applicability.

Research Approach

The chosen research approach is deductive. This approach begins with existing theories and empirical studies on employee attrition and machine learning, forming hypotheses that are tested using data-driven methodologies. Based on literature review findings, hypotheses are constructed regarding the factors influencing employee attrition, such as “employees with low work-life balance scores are more likely to leave the organization.” Statistical and machine learning methods are applied to test these hypotheses and validate their significance. Predictive models are developed, with their performance evaluated against predefined metrics to ensure robustness.

Although primarily deductive, this research incorporates inductive elements through exploratory data analysis (EDA). By identifying unexpected patterns or correlations in the data, EDA provides insights that may not have been anticipated in the hypothesis formulation stage.

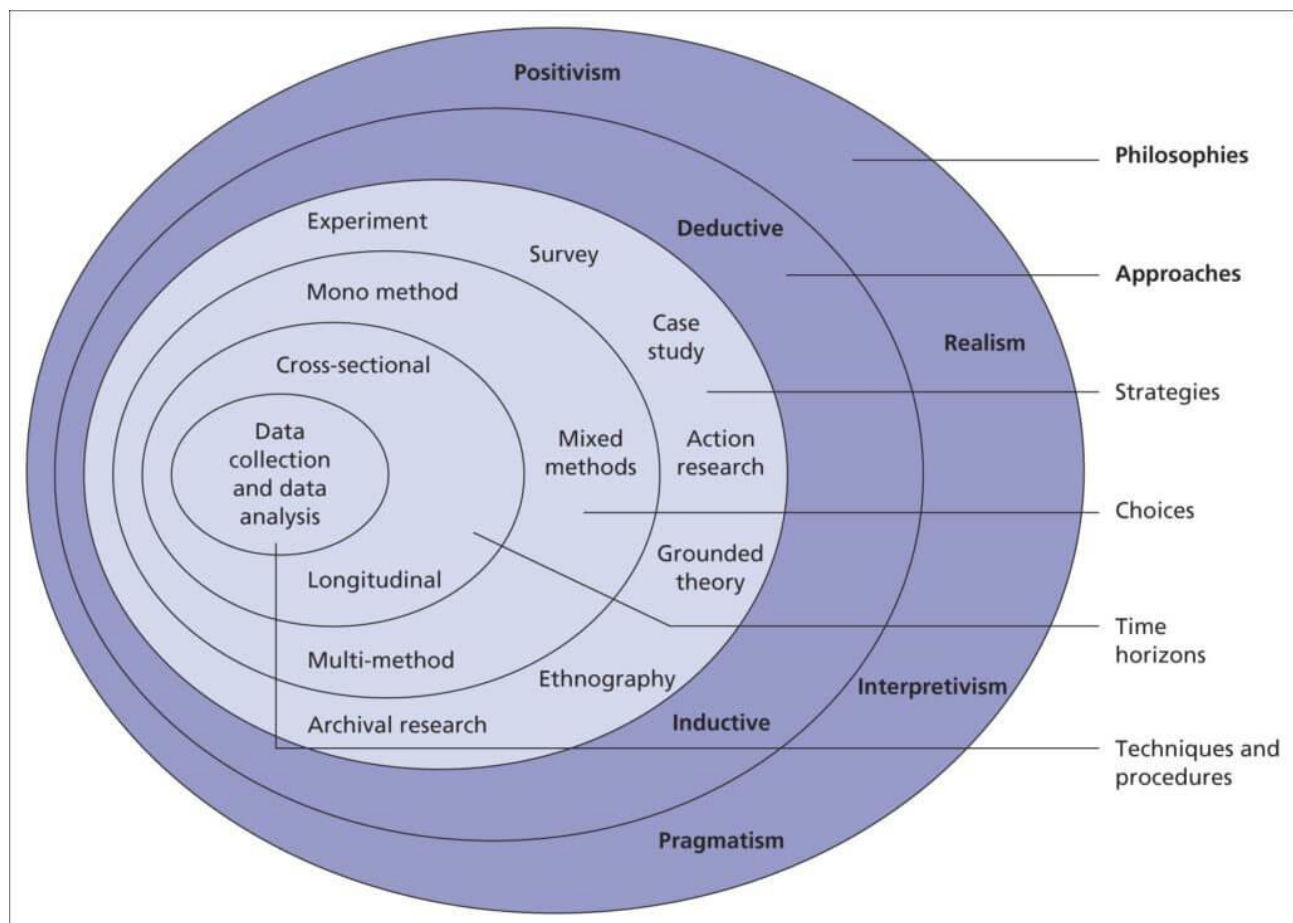


Fig: Research Onion Framework

Source:(Saunders et al., 2012)

Research Philosophy

The research philosophy underpinning this study is positivism. Positivism is predicated on the belief that knowledge is derived from observable phenomena and measurable data. In this context, employee attrition is examined through empirical data and statistical analysis to uncover patterns and derive conclusions.

This philosophy is appropriate as the dataset provides concrete, quantifiable data that supports the development and testing of hypotheses through machine learning models. Positivism ensures an unbiased analysis by emphasizing objectivity in data collection, preprocessing, and model evaluation. The findings, grounded in statistical validity and machine learning techniques, can potentially be generalized to similar organizational contexts. While interpretivism or pragmatism could offer a nuanced understanding of individual employee experiences, they are less suitable for this study due to the objective, data-centric nature of the research.

Methodological Framework

The methodological framework integrates various components to ensure a systematic and comprehensive analysis. The framework adheres to the CRISP-DM (Cross-Industry Standard Process for Data Mining) model, widely used in data science projects. The first stage focuses on business understanding, clearly defining the problem of employee attrition and its organizational impact. This is followed by data understanding, involving detailed exploration of the dataset to identify key variables, missing values, and potential biases. During data preparation, preprocessing steps such as handling missing values, encoding categorical variables, and normalizing numerical features are undertaken.

Modelling involves applying supervised learning techniques, including Logistic Regression, Decision Tree, Random Forest, K-NN, SVM, Kernel SVM, Naive Bayes, Gradient Boosting, and ADA Boosting. The evaluation phase compares model performances using metrics such as confusion matrices to determine their effectiveness. Although outside the scope of this dissertation, practical deployment scenarios and potential integration with HR systems are discussed in later chapters.

RESEARCH PLAN (FLOWCHART)

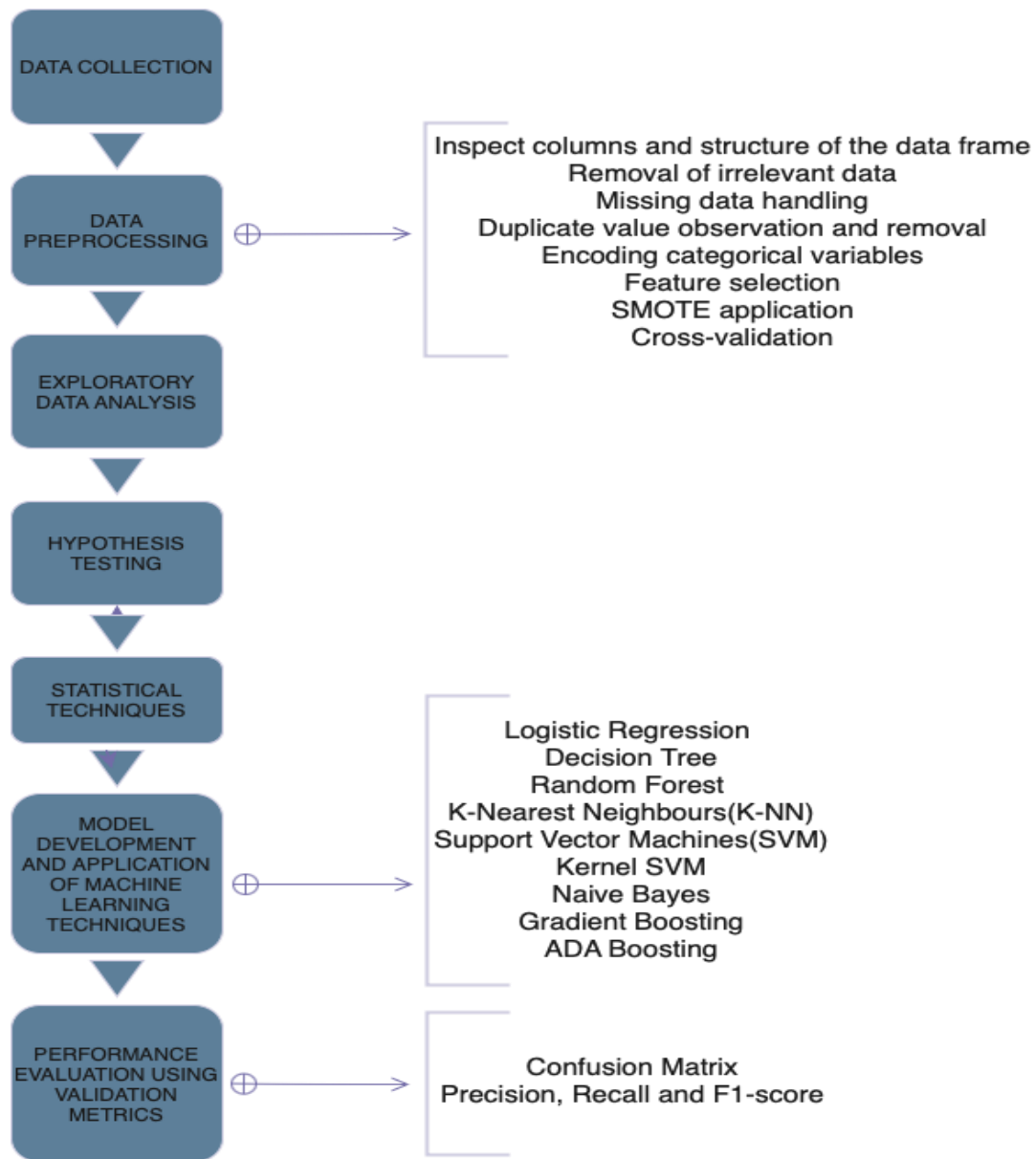


Fig 1: Research Plan

DATA COLLECTION

Dataset Overview

The dataset used for this study, sourced from Kaggle, is titled “Capstone Project: IBM Employee Attrition Prediction.” Created by IBM data scientists, this hypothetical dataset contains 23,436 rows and 37 columns with a mix of numeric and categorical variables. It serves as a labelled dataset for supervised learning, where the target variable, Attrition, indicates whether an employee has left the organization (Yes = 1, No = 0). The dataset provides a comprehensive view of employee demographics, job characteristics, and performance metrics. This wealth of information enables the development of machine learning models that can identify at-risk employees, providing HR teams with actionable insights for retention strategies. The dataset can be accessed [here](#).

Business Problem

High employee attrition disrupts business operations and incurs significant costs associated with hiring and training new staff. This research focuses on identifying the factors that contribute to employee attrition and developing predictive models to address it. The goals are to:

- Understand the key factors influencing attrition through **Exploratory Data Analysis (EDA)**.
- Build machine learning models to predict at-risk employees.
- Equip HR professionals with insights to proactively address retention challenges.

The success of this approach depends not only on the accuracy of predictive models but also on the human element—engaging with employees to understand their concerns and taking steps to address controllable factors.

Feature Description

The dataset comprises 37 features, each contributing to the understanding of employee behaviour and attrition. Below is a detailed explanation of these features:

Feature	Description	Type
Age	Numerical representation of the employee's age, capturing career stage and retirement proximity.	Numeric
Attrition	Target variable indicating whether the employee left the organization (Yes = 1, No = 0).	Categorical
BusinessTravel	Frequency of business travel categorized as No Travel, Travel Rarely, and Travel Frequently.	Categorical
DailyRate	Represents the daily salary rate of the employee, highlighting compensation variations.	Numeric
Department	Department where the employee works, such as HR, R&D, or Sales, indicating functional alignment.	Categorical
DistanceFromHome	The distance (in kilometers) between the employee's residence and the workplace, often correlating with satisfaction levels.	Numeric

Education	Education level on a scale (e.g., High School = 1, Bachelor = 2), reflecting academic qualifications.	Numeric
EducationField	Employee's field of study, including Life Sciences, Marketing, and Technical Degree, indicating specialization.	Categorical
EnvironmentSatisfaction	Satisfaction with the work environment, rated on a scale of 1 to 4, affecting engagement and productivity.	Numeric
Gender	Gender of the employee (Male/Female).	Categorical
HourlyRate	Hourly wage of the employee, useful for assessing compensation fairness.	Numeric
JobInvolvement	Extent of the employee's involvement in their role, rated from 1 (low) to 4 (high).	Numeric
JobLevel	Hierarchical level of the job (e.g., entry-level, mid-level), indicating seniority and career progression.	Numeric
JobRole	Specific job designation, such as Manager, Sales Executive, or Research Scientist, reflecting role diversity.	Categorical
JobSatisfaction	Employee's satisfaction with their job, rated on a scale of 1 to 4.	Numeric
MaritalStatus	Marital status categorized as Single, Married, or Divorced, potentially influencing work-life dynamics.	Categorical
MonthlyIncome	Employee's total monthly salary, highlighting compensation.	Numeric
OverTime	Indicates whether the employee works overtime (Yes/No), often correlating with job stress and attrition.	Categorical
PerformanceRating	Performance appraisal score (1 to 4), indicating efficiency and contribution.	Numeric
RelationshipSatisfaction	Satisfaction with interpersonal relationships at work, rated from 1 to 4.	Numeric

StockOptionLevel	Stock options provided, reflecting organizational incentives for retention.	Numeric
TotalWorkingYears	Total professional experience in years, indicating career maturity.	Numeric
TrainingTimesLastYear	Number of training sessions attended, indicating opportunities for skill development.	Numeric
WorkLifeBalance	Perceived balance between work responsibilities and personal life, rated from 1 to 4.	Numeric
YearsAtCompany	Number of years spent in the current organization, indicating organizational loyalty.	Numeric
YearsInCurrentRole	Tenure in the current role, reflecting job stability and growth.	Numeric
YearsSinceLastPromotion	Years since the employee's last promotion, highlighting career advancement.	Numeric
YearsWithCurrManager	Number of years under the current manager, reflecting team dynamics and satisfaction.	Numeric

Table 1: Feature Description

- **Age:**
The age distribution shows a range from 18 to 60 years, with most employees falling in the 30–40 age bracket. This indicates a workforce dominated by mid-career professionals.
- **Attrition:**
The attrition variable indicates that approximately 16% of employees have left the organization, highlighting a class imbalance. The remaining 84% are current employees.
- **BusinessTravel:**
The majority of employees (71%) travel rarely for business. Employees traveling frequently account for 19%, while 10% do not travel at all.
- **DailyRate:**
Daily income rates are evenly distributed across the dataset, ranging from 102 to 1,499. This suggests a fair spread of compensation levels.

- **Department:**

Most employees (65%) work in the Research & Development department. Sales account for 30%, while Human Resources and other departments make up the remaining 5%.

- **DistanceFromHome:**

The majority of employees live within 10 kilometres of their workplace, with distances ranging up to 29 kilometres. Proximity to the workplace could influence satisfaction levels.

- **Education:**

Educational levels range from 1 (High School) to 5 (Doctorate), with the majority of employees holding a Bachelor's or Master's degree.

- **EducationField:**

Life Sciences dominate the dataset (41%), followed by Medical fields (31%). Other fields like Marketing, Technical, and Humanities make up the remaining proportion.

- **EmployeeCount:**

This feature is a constant value of 1 for all employees and is not relevant for predictive analysis.

- **EmployeeNumber:**

Employee numbers are unique identifiers for each record and are not used for model training.

- **EnvironmentSatisfaction:**

Ratings range from 1 (low) to 4 (high), with the majority of employees reporting high satisfaction levels (3–4).

- **Gender:**

The dataset shows a male-to-female ratio of 60:40, suggesting a relatively balanced gender distribution.

- **HourlyRate:**

Hourly rates range from 30 to 100, with a fairly uniform distribution, indicating consistent hourly compensation across roles.

- **JobInvolvement:**

Most employees report high involvement levels, with ratings predominantly at 3 or 4, reflecting strong engagement.

- **JobLevel:**

The majority of employees fall within levels 1 to 3, indicating a large representation of early- and mid-career roles.

- **JobRole:**

Roles are diverse, with Sales Executives (22%) and Research Scientists (20%) being the most common. Other roles like Managers and Laboratory Technicians make up the rest.

- **JobSatisfaction:**

Satisfaction levels are high, with the majority rating their satisfaction as 3 or 4, suggesting overall contentment in job roles.

- **MaritalStatus:**

Married employees constitute 46% of the dataset, followed by single (32%) and divorced (22%) employees.

- **MonthlyIncome:**

Monthly incomes range from 1,009 to 20,000, with a skewed distribution towards lower income levels.

- **MonthlyRate:**

Monthly rates vary widely, from 2,094 to 27,000, reflecting significant differences in compensation structures.

- **NumCompaniesWorked:**

Employees have worked for 0 to 9 companies, with many having 1–3 prior employers, indicating a moderately stable workforce.

- **Over18:**

This feature is a constant with a value of “Yes” for all employees, making it irrelevant for analysis.

- **OverTime:**

Around 28% of employees work overtime, which could correlate with stress or high workload, while 72% do not.

- **PercentSalaryHike:**

Salary hikes range from 11% to 25%, with the majority receiving moderate increases between 15% and 20%.

- **PerformanceRating:**

Nearly all employees are rated at 3 or 4, suggesting a high-performing workforce.

- **RelationshipSatisfaction:**

Ratings range from 1 to 4, with most employees rating their satisfaction as 3 or 4, reflecting positive interpersonal relationships.

- **StandardHours:**

This feature is a constant value of 80 for all employees, providing no variability for analysis.

- **StockOptionLevel:**

Stock option levels range from 0 to 3, with the majority of employees at level 0, indicating limited stock incentives.

- **TotalWorkingYears:**

Experience ranges from 0 to 40 years, with the majority of employees having 5–15 years of total work experience.

- **TrainingTimesLastYear:**

Employees attended 0 to 6 training sessions, with most participating in 2–3 sessions, indicating moderate training opportunities.

- **WorkLifeBalance:**

Most employees rate their work-life balance as 3 or 4, suggesting overall satisfaction with their work-life integration.

- **YearsAtCompany:**

Tenure at the current company ranges from 0 to 40 years, with the majority of employees having worked 3–10 years.

- **YearsInCurrentRole:**

Employees have spent 0 to 22 years in their current role, with the majority having 2–5 years of role-specific experience.

- **YearsSinceLastPromotion:**

Time since the last promotion ranges from 0 to 15 years, with many employees having recent promotions (0–3 years).

- **YearsWithCurrManager:**

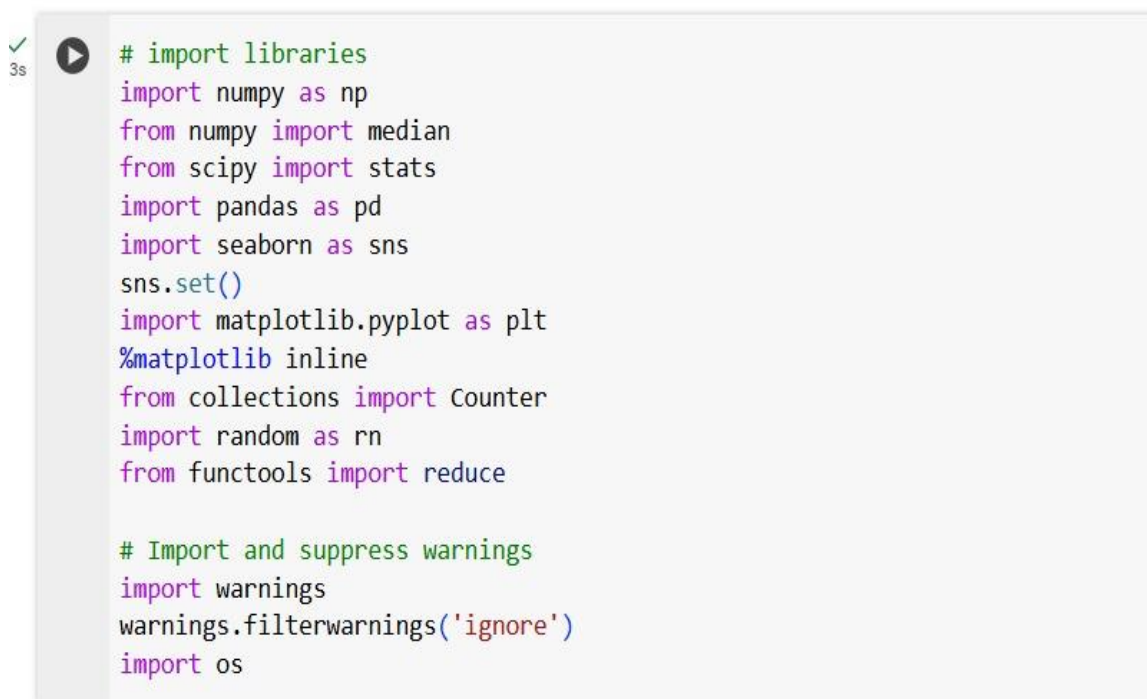
Employees have worked with their current manager for 0 to 17 years, with most reporting 2–5 years of managerial continuity.

Data Preprocessing

Data preprocessing is a foundational step in preparing raw data for analysis and modelling. It ensures the data is clean, consistent, and structured to enhance the accuracy and reliability of predictive models. In this project, preprocessing involved handling a dataset with 23,436 rows and 37 columns, which was reduced to 23,141 rows and 32 columns after cleaning. The following section details each preprocessing technique used, explaining its purpose, implementation, and relevance to the project.

Importing Libraries

To initiate preprocessing, essential Python libraries were imported. **Pandas** was used for data manipulation, enabling efficient handling of tabular data. **NumPy** facilitated numerical computations, while **Matplotlib** and **Seaborn** were employed for visualization, assisting in identifying patterns and anomalies. Additionally, **Scikit-learn** provided tools for encoding categorical variables and scaling numerical features. Suppressing unnecessary warnings using the **Warnings** library improved the readability of outputs during execution. These libraries formed a comprehensive toolkit for preprocessing, ensuring robust data handling and visualization.



```
# import libraries
import numpy as np
from numpy import median
from scipy import stats
import pandas as pd
import seaborn as sns
sns.set()
import matplotlib.pyplot as plt
%matplotlib inline
from collections import Counter
import random as rn
from functools import reduce

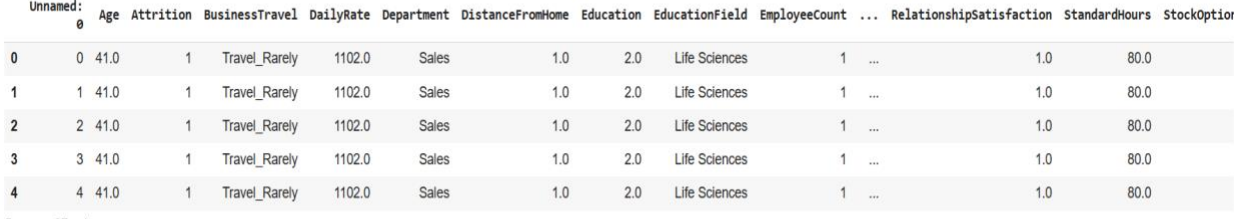
# Import and suppress warnings
import warnings
warnings.filterwarnings('ignore')
import os
```

Fig 2: Importing Libraries

Loading the Dataset

The dataset was loaded into a Pandas DataFrame using the `pd.read_csv()` function. This step is crucial as it integrates the raw CSV file into a manipulatable format. Initial inspection of the dataset revealed 37 features comprising numerical and categorical variables. Commands such as `df.info()` and `df.describe()` provided insights into data types, ranges, and distributions. This early analysis identified potential issues, such as irrelevant features and constant columns, which were addressed in subsequent steps.

```
[4] # Read the file
df = pd.read_csv('IBM HR Data new.csv')
df.head()
```



	Unnamed: 0	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	...	RelationshipSatisfaction	StandardHours	StockOption
0	0	41.0	1	Travel_Rarely	1102.0	Sales	1.0	2.0	Life Sciences	1	...	1.0	80.0	
1	1	41.0	1	Travel_Rarely	1102.0	Sales	1.0	2.0	Life Sciences	1	...	1.0	80.0	
2	2	41.0	1	Travel_Rarely	1102.0	Sales	1.0	2.0	Life Sciences	1	...	1.0	80.0	
3	3	41.0	1	Travel_Rarely	1102.0	Sales	1.0	2.0	Life Sciences	1	...	1.0	80.0	
4	4	41.0	1	Travel_Rarely	1102.0	Sales	1.0	2.0	Life Sciences	1	...	1.0	80.0	

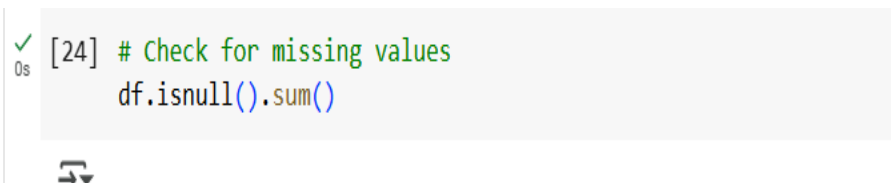
5 rows x 37 columns

Fig 3: Loading the Dataset

Handling Missing Values

One of the critical steps in data preprocessing is identifying and handling missing values, as they can significantly impact the performance and accuracy of machine learning models. In this dataset, I began by verifying the presence of missing values using the `df.isnull().sum()` method, which calculates the total number of missing entries for each column. The dataset used for final analysis had no missing values, meaning all columns were fully populated after the data cleaning process.

```
✓ [24] # Check for missing values
0s df.isnull().sum()
```





	0
Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobRole	0
JobSatisfaction	0
MaritalStatus	0
MonthlyIncome	0
MonthlyRate	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0
RelationshipSatisfaction	0
StandardHours	0
StockOptionLevel	0
TotalWorkingYears	0
TrainingTimesLastYear	0
WorkLifeBalance	0
YearsAtCompany	0
YearsInCurrentRole	0
YearsSinceLastPromotion	0
YearsWithCurrManager	0

Fig 4: Handling missing values

Removing Duplicates

Duplicate rows are records that contain identical values across all columns. They often arise due to data entry errors or redundancies during data collection. Removing duplicates ensures the dataset represents unique observations, preventing overrepresentation of specific entries, which could bias the analysis or model outcomes. In this project, duplicates were identified using the `df.duplicated()` function, and 15 duplicate rows were removed using the `df.drop_duplicates()` method. This reduced the dataset to 23,421 rows, ensuring each record contributed unique information to the analysis.

```
[25] # Check for duplicate observations
df.duplicated().sum()
```

15

```
duplicate_rows = df[df.duplicated()]
print("\nDuplicate Rows:")
duplicate_rows.head()
```

Duplicate Rows:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	Environment
10	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
11	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
12	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	7	
660	37	1	Travel_Rarely	807	Human Resources	6	4	Human Resources	1		1
661	37	1	Travel_Rarely	807	Human Resources	6	4	Marketing	1		4

Fig 5: Removing Duplicate Records

Dropping Irrelevant Features

Irrelevant features are columns that provide no meaningful information for analysis. For instance, columns like EmployeeCount, Over18, and StandardHours were constant, offering no variability. Additionally, EmployeeNumber, a unique identifier, was excluded as it lacked analytical value. Removing these columns streamlined the dataset, reducing noise and enhancing computational efficiency. This step was essential for focusing the analysis on features directly influencing employee attrition.

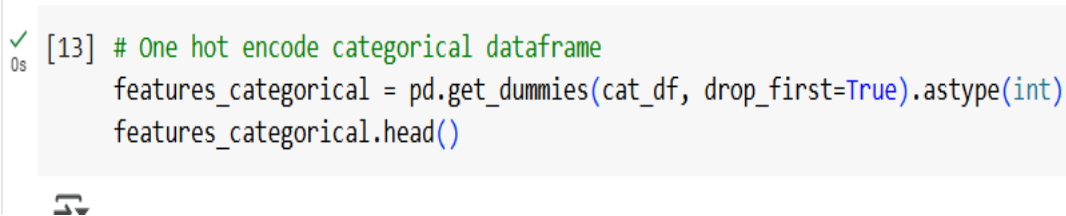
```
[ ] df2 = df3.drop(labels = ['Over18', 'StandardHours', 'EmployeeCount'], axis = 1)
```

Fig 6: Dropping irrelevant features

Encoding Categorical Variables

Categorical variables must be converted into numerical formats to be utilized by machine learning algorithms. In this project, **label encoding** was applied to binary variables like Gender, assigning numerical values (e.g., 0 for Male, 1 for Female). **One-hot encoding** was used for multi-class

variables like Department and JobRole, creating separate binary columns for each category. This transformation preserved the informational value of categorical variables while making them compatible with predictive models. Encoding was crucial for ensuring that the categorical data contributed meaningfully to the analysis without introducing biases.



```
[13] # One hot encode categorical dataframe
      features_categorical = pd.get_dummies(cat_df, drop_first=True).astype(int)
      features_categorical.head()
```

Fig 7: Encoding categorical variables

Handling Class Imbalance

Class imbalance occurs when the target variable has an unequal distribution of categories. In this dataset, the Attrition variable had significantly more "No" labels than "Yes" labels, creating an imbalance that could bias the model toward the majority class. To address this, the **Synthetic Minority Oversampling Technique (SMOTE)** was applied. SMOTE generates synthetic samples for the minority class, balancing the dataset and improving the model's ability to learn patterns associated with attrition. This step was critical for ensuring fair and accurate predictions.

Scaling and Normalization

Numerical features often vary significantly in magnitude, which can disproportionately influence machine learning models. Scaling ensures all numerical features contribute equally, enhancing model performance. For this project, Min-Max Scaling was applied to features such as Age, DistanceFromHome, and MonthlyIncome. This transformation rescaled values to a range between 0 and 1, promoting consistency across features and improving the efficiency of gradient-based optimization algorithms during model training.

Final Dataset Preparation

After completing the preprocessing steps, the final dataset contained 23,141 rows and 32 columns. The target variable, Attrition, was moved to the last column to improve clarity during model training. A final inspection using commands like `df.info()` confirmed that the dataset was clean, balanced, and ready for analysis. This organized structure facilitated seamless integration with machine learning pipelines.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a pivotal phase in any data-driven project, serving as a means to explore, summarize, and visualize the dataset. This step not only helps uncover patterns and trends but also ensures that the data aligns with the objectives of the study. In this project, EDA was performed on the cleaned dataset comprising 23,139 rows and 32 features, both numerical and categorical. The primary aim of this analysis was to understand the factors contributing to employee attrition and identify patterns that could inform predictive modelling.

The dataset's target variable, Attrition, was analysed alongside other features to assess their relationships and contributions to the overall attrition rate. A combination of descriptive statistics, feature-specific analyses, and visualizations were employed to gain a comprehensive understanding of the data. This section provides an overview of the EDA process, detailing the methodologies applied to individual features and their implications.

Analysis of the Target Variable: Attrition

The target variable, Attrition, is a binary categorical feature indicating whether an employee has left the company (1 or Yes) or is still employed (0 or No). After data cleaning, the dataset contains 23,139 employee records and 32 features. An analysis of Attrition revealed that **3,653 employees (15.79%)** have left the company, while **19,486 employees (84.21%)** continue to work. This highlights a notable class imbalance, with the majority of employees classified as "Currently Working in the Company." Such imbalance is crucial to address during the modelling phase, as it may lead to biases favouring the majority class.

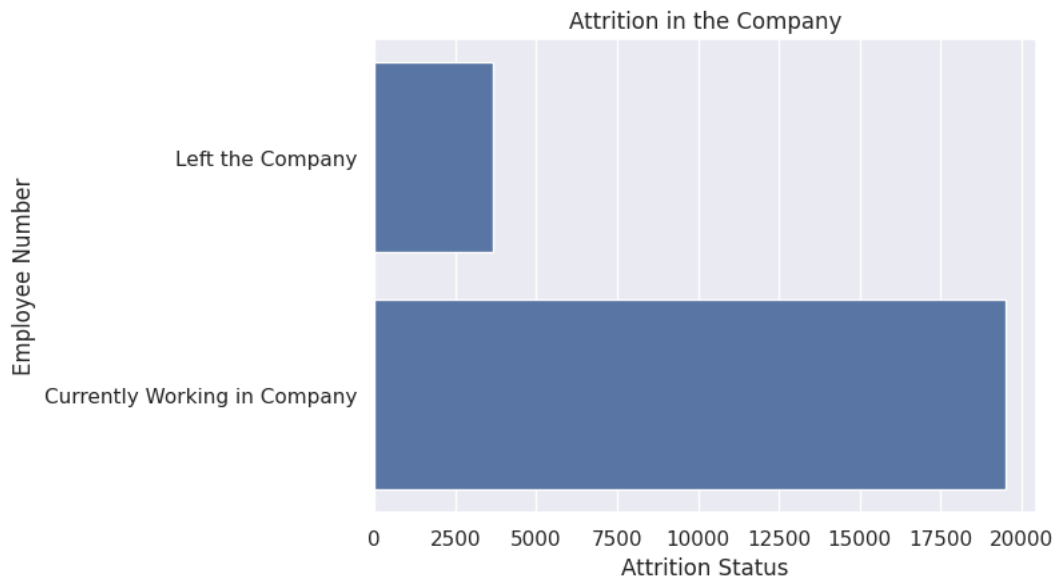


Fig 8: Distribution of Attrition in the company

Exploration of the Features

Age

```
[101] # Age Group & Attrition Table
      table_df(x = 'Age')
```

Age	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0 (18-21)	639	341	53.364632	9.334793
1 (22-25)	1306	349	26.722818	9.553791
2 (26-35)	9507	1756	18.470601	48.070079
3 (36-60)	11687	1207	10.327715	33.041336

Fig 9: Attrition compared to age group

The analysis of the Age feature revealed significant trends in attrition across four age groups: 18-21, 22-25, 26-35, and 36-60. Employees in the youngest age group (18-21) had the highest attrition proportion within their group, with **53.4% (341 out of 639)** leaving the company, accounting for **9.3% of overall attrition cases**. This suggests that younger employees are more likely to leave, possibly due to career exploration. In contrast, the 26-35 age group recorded the highest absolute attrition numbers, with **1,756 employees leaving out of 9,507**, contributing to **48% of total attrition**. This group, often comprising mid-career professionals, may leave due to dissatisfaction with career progression or work-life balance. Employees aged 36-60 exhibited the lowest attrition rates, likely valuing job stability and security as they advance in their careers.

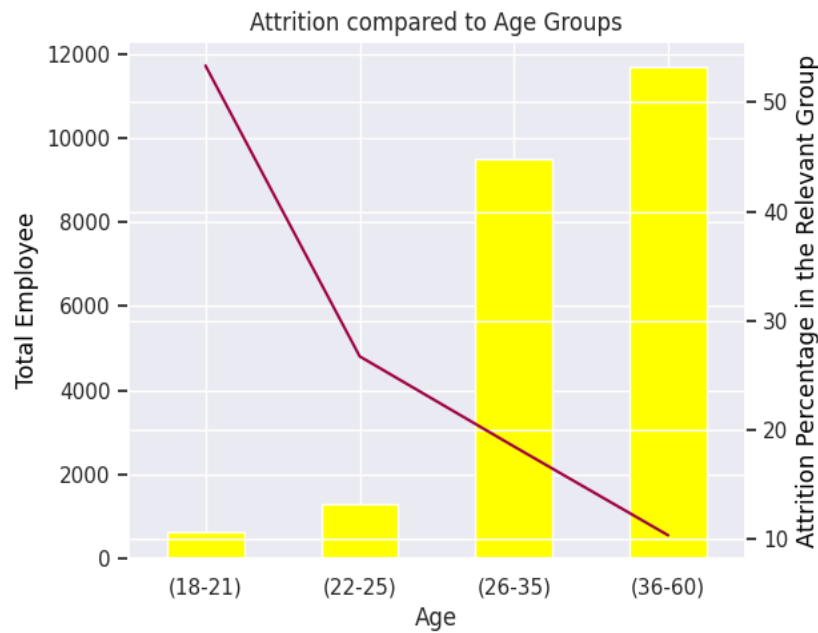


Fig 10: Graph showing attrition compared to age group

Business Travel

```
[103] # Business Travel & Attrition Table
table_df(x = 'BusinessTravel')
```

	BusinessTravel	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Non-Travel	2362	187	7.917019	5.119080
1	Travel_Frequently	4367	1080	24.730937	29.564741
2	Travel_Rarely	16410	2386	14.539915	65.316178

Fig 11: Attrition compared to business travel

The Business Travel feature categorizes employees based on travel frequency: Non-Travel, Travel_Rarely, and Travel_Frequently. A significant majority of employees, **81.13% (18,772 out of 23,139)**, fall under the Non-Travel or Travel_Rarely groups. Among them, the Travel_Rarely group recorded the highest attrition numbers, with **2,386 employees leaving**, constituting **14.5% of this group** and **65.3% of overall attrition**. Employees in the Travel_Frequently group, comprising **18.87% of employees (4,367 out of 23,139)**, displayed the highest attrition proportion within their group, at **24.7% (1,080 out of 4,367)**, contributing to **29.5% of total attrition**.

Conversely, employees in the Non-Travel group had the lowest attrition rates, with only **7.9% of the group** leaving the company. These findings indicate that frequent travel obligations increase the

likelihood of attrition, suggesting that organizations should focus on addressing challenges faced by employees in travel-intensive roles to improve retention.

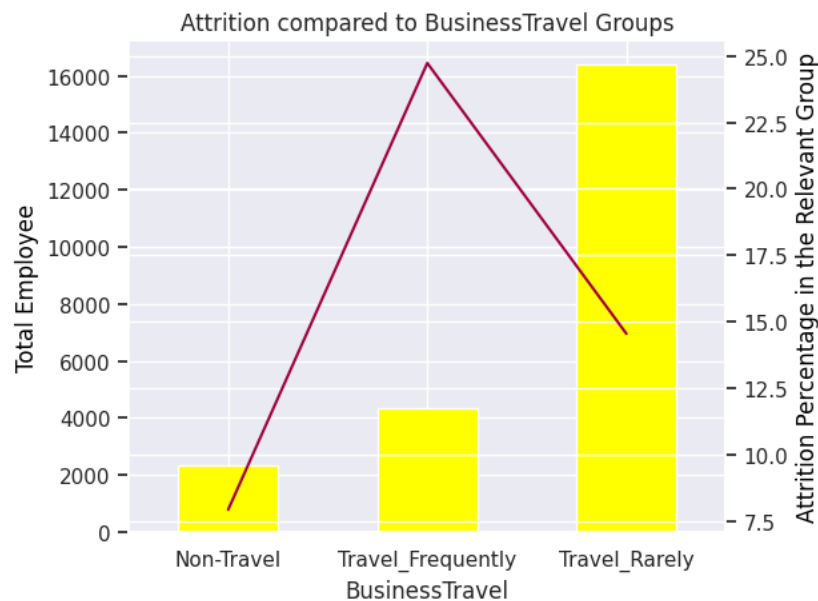


Fig 12: Graph showing attrition compared to business travel

DEPARTMENT

```
[105] # Department & Attrition Table
table_df(x = 'Department')
```

	Department	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Human Resources	1003	187	18.644068	5.119080
1	Research & Development	15097	2036	13.486123	55.735012
2	Sales	7039	1430	20.315386	39.145907

Fig 13: Attrition compared to department

The **Department** feature categorizes employees into three groups: **Human Resources**, **Research & Development**, and **Sales**. Analysing the attrition trends within these departments revealed distinct patterns, highlighting the departments most affected by employee turnover.

The **Research & Development** department has the largest workforce and recorded the highest number of attrition cases in absolute terms, with 2,036 employees leaving the company, accounting for 55.7% of overall attrition. However, as a percentage of its own group, the attrition rate for

Research & Development is the lowest at 13.4%, suggesting that its larger workforce dilutes its relative attrition impact.

In contrast, the Sales department experienced the highest attrition rate within its group, with **20.3% of employees leaving** (1,430 out of 7,039). This accounts for **39.1% of overall attrition**, indicating that employees in sales roles may face challenges contributing to higher turnover. The Human Resources department follows, with an attrition rate of **18.6%**, equating to **187 employees leaving out of 1,003**, but this represents only **5.1% of overall attrition**.

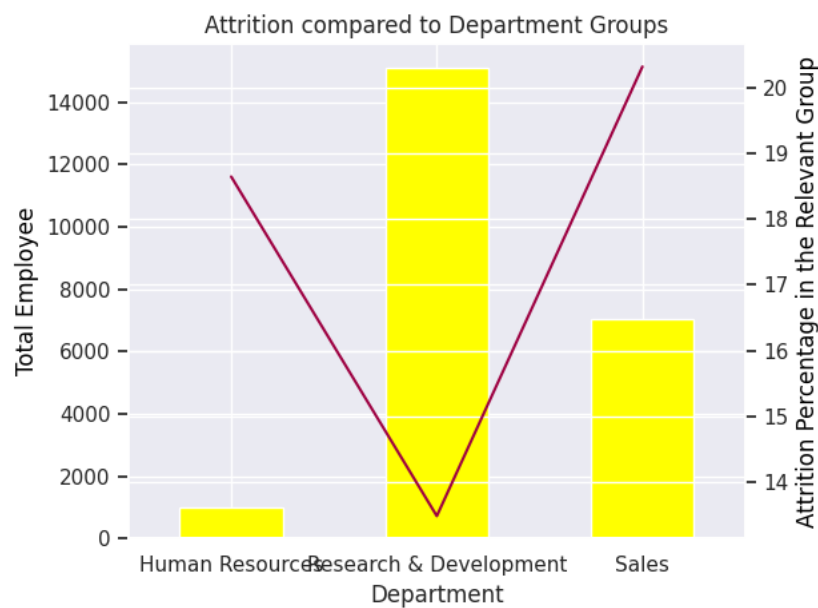


Fig 14: Graph showing attrition compared to department

Education

```
[111] # Education Level & Attrition Table
table_df(x = 'Education')
```

	Education	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Bachelor	8984	1503	16.729742	41.144265
1	Below College	2697	496	18.390805	13.577881
2	College	4465	687	15.386338	18.806460
3	Doctor	753	79	10.491368	2.162606
4	Master	6240	888	14.230769	24.308787

Fig 15: Attrition compared to education

The Education feature categorizes employees based on their highest level of education, revealing key attrition trends. Employees with a Bachelor's degree experienced the highest attrition, with **1,503 employees leaving**, accounting for **41.1% of overall attrition**. This is likely due to their large representation in the company. In contrast, employees with a Doctorate degree had the lowest attrition, with only **79 employees leaving**, representing **2.16% of total attrition**, indicating greater stability or satisfaction in their roles.

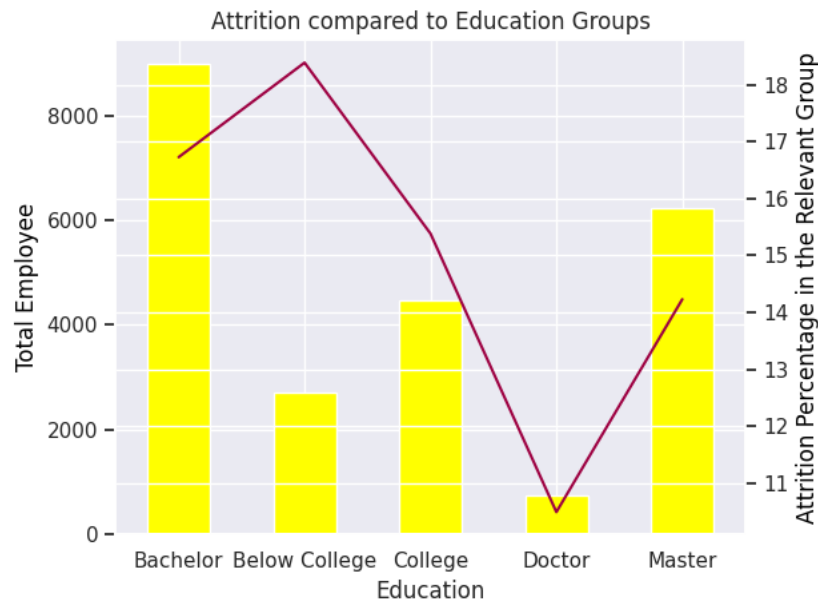


Fig 16: Graph showing attrition compared to education

Gender

```
[118] # Gender & Attrition Table
table_df(x = 'Gender')
```

	Gender	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Female	9257	1416	15.296532	38.762661
1	Male	13882	2237	16.114393	61.237339

Fig 17: Attrition compared to gender

The analysis of the Gender feature revealed that male employees are more likely to leave the company than female employees. This pattern suggests that factors influencing attrition may disproportionately impact male employees or that they may be more inclined to seek external opportunities.

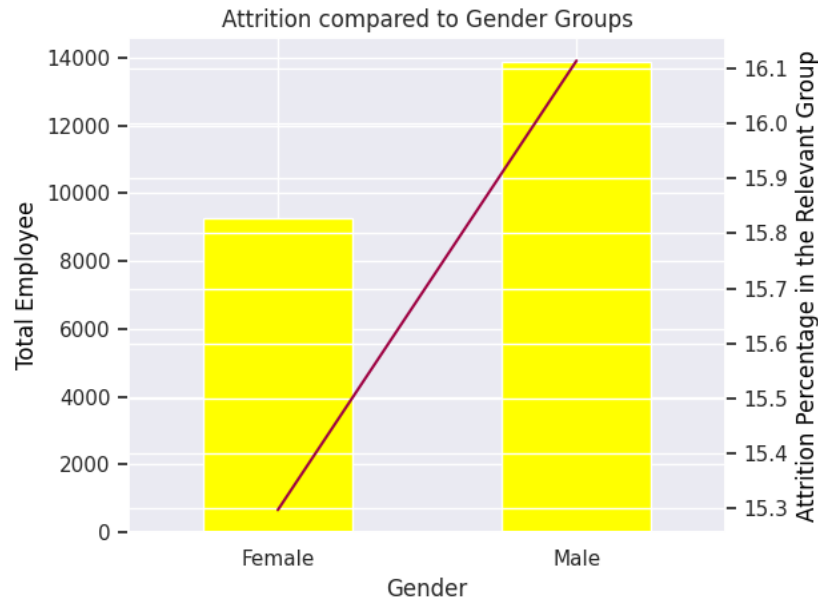


Fig 18: Graph showing attrition compared to gender

Job Role

```
[126] # Job Role & Attrition Table
table_df(x = 'JobRole')
```

	JobRole	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Healthcare Representative	2072	289	13.947876	7.911306
1	Human Resources	835	148	17.724551	4.051465
2	Laboratory Technician	4097	748	18.257261	20.476321
3	Manager	1584	138	8.712121	3.777717
4	Manufacturing Director	2357	287	12.176496	7.856556
5	Research Director	1279	124	9.695074	3.394470
6	Research Scientist	4580	738	16.113537	20.202573
7	Sales Executive	5036	817	16.223193	22.365179
8	Sales Representative	1299	364	28.021555	9.964413

Fig 19: Attrition compared to job role

The **Job Role** feature provides insights into attrition trends across various positions in the organization. **Sales Executives** experienced the highest attrition, with 817 employees leaving, representing 22.3% of total attrition. **Laboratory Technicians** and **Research Scientists** followed closely, contributing 748 and 738 attrition cases, respectively, making these three roles account for 40.6% of all attrition.

The Sales Representative role, despite a smaller workforce, had the highest attrition rate within its group, with **28% of employees leaving**, highlighting significant retention challenges. In contrast, the Manager role demonstrated the lowest attrition, with only **138 employees leaving** (8.7% of total attrition and 3.7% within its group). These findings emphasize the need for targeted retention strategies, particularly for high-turnover roles, to improve job satisfaction, provide growth opportunities, and reduce overall attrition.

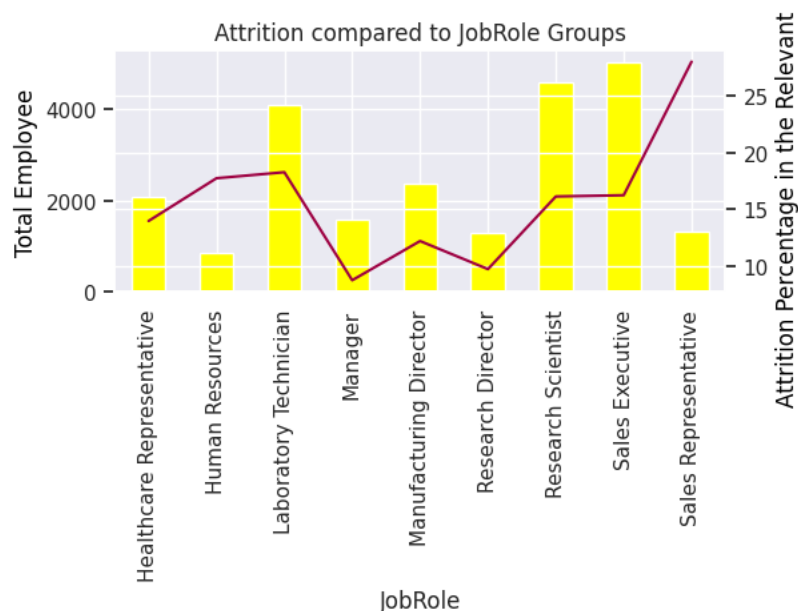


Fig 20: Graph showing attrition compared to job role

Job Satisfaction

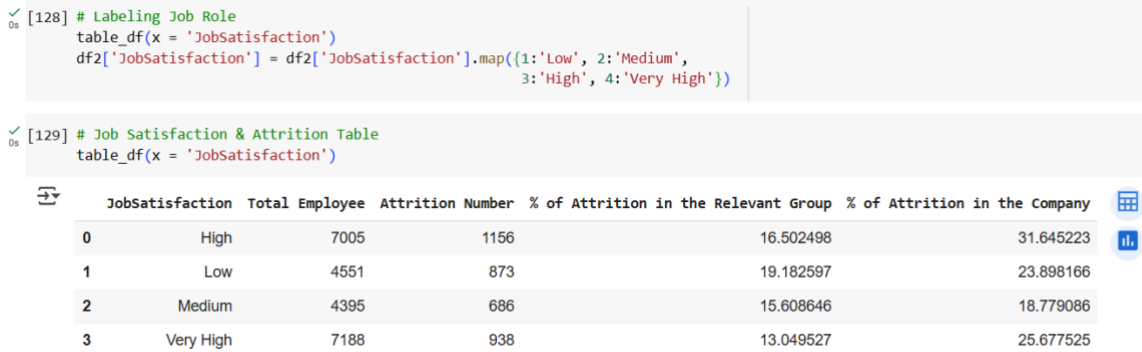


Fig 21: Attrition compared to job satisfaction

The analysis of the **Job Satisfaction** feature offers intriguing insights into the dynamics of employee attrition. Employees with high job satisfaction surprisingly account for the largest proportion of attrition, contributing to 31.6% of the company's total attrition. This finding contradicts the assumption that high job satisfaction alone ensures employee retention. It suggests that while job satisfaction is a significant factor, other elements, such as career growth opportunities, work-life balance, or compensation, might play a more substantial role in influencing attrition. On the other hand, low job satisfaction is also a notable contributor to attrition, comprising 23.9% of total attrition in the company.

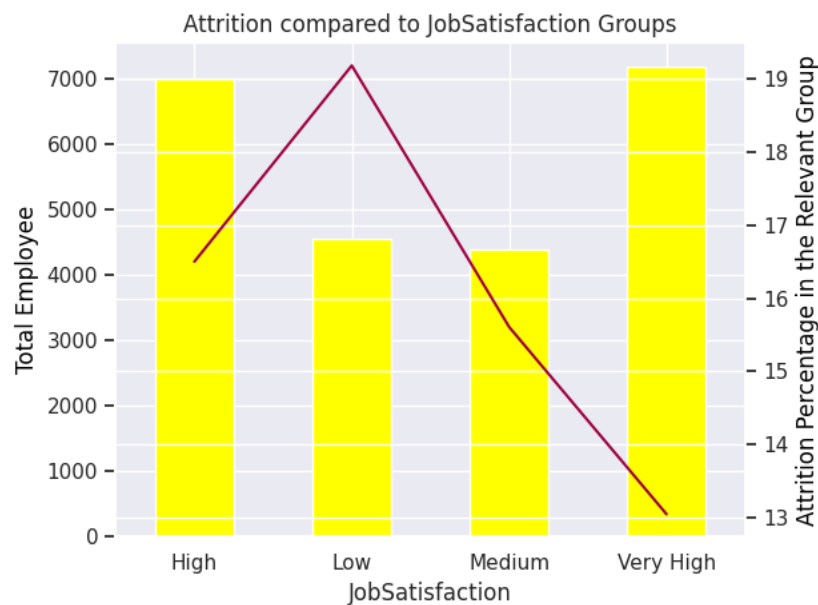


Fig 22: Graph showing attrition compared to job satisfaction

Monthly Income

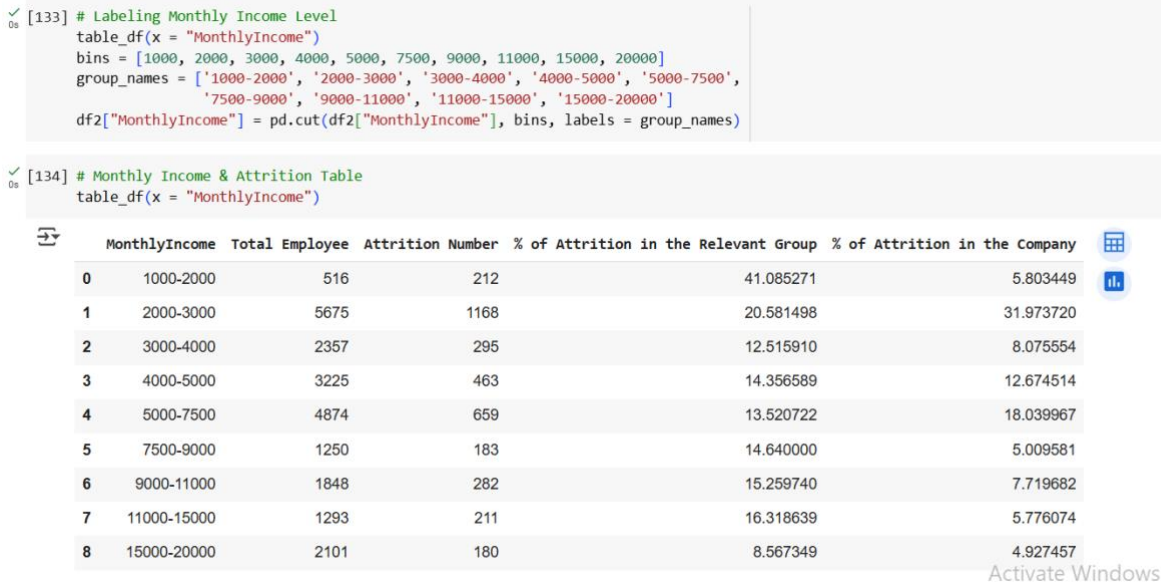


Fig 23: Attrition compared to monthly income

The analysis of the Monthly Income feature reveals a clear relationship between income levels and employee attrition. Employees earning between **\$2000-\$3000** per month exhibit the highest attrition rate, contributing to **31.9% of overall attrition**. This trend suggests dissatisfaction or limited financial motivation in this income group. Interestingly, employees earning **\$1000-\$2000** monthly demonstrate the highest attrition proportion within their income group, with **41.1% leaving the company**, potentially reflecting financial constraints driving turnover.

A general decline in attrition is observed as income increases, supporting the idea that higher salaries enhance employee retention. However, an anomaly is noted in the **\$9000-\$11,000** income bracket, where attrition rises within its group to **15.3%**, indicating possible dissatisfaction despite higher earnings.

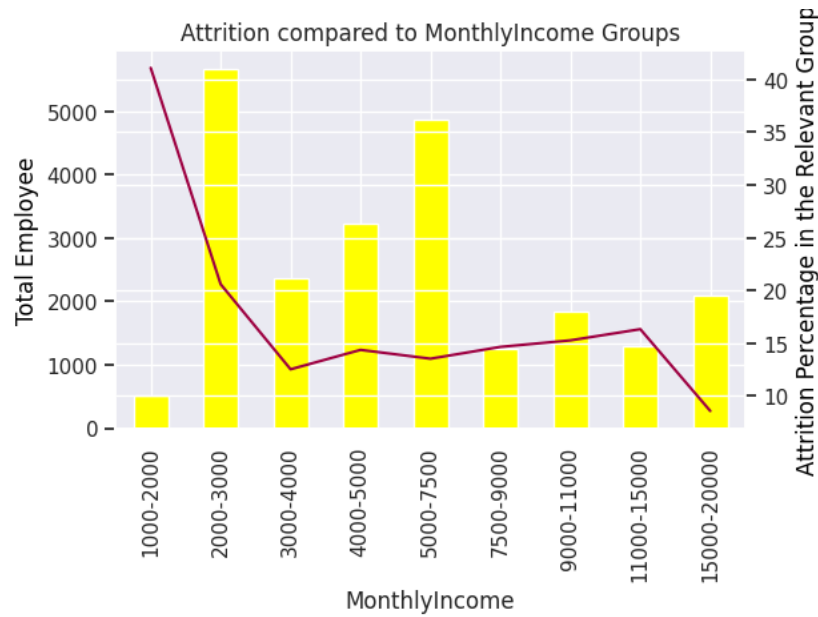


Fig 24: Graph showing attrition compared to monthly income

Number of Companies Worked

```
[137] # Numbers Companied Worked & Attrition Table
table_df(x = 'NumCompaniesWorked')
```

	NumCompaniesWorked	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	0	3138	446	14.212874	12.209143
1	1	8195	1317	16.070775	36.052560
2	2	2298	326	14.186249	8.924172
3	3	2489	364	14.624347	9.964413
4	4	2176	276	12.683824	7.555434
5	5+	4843	924	19.079083	25.294279

Fig 25: attrition compared to number of companies worked

The feature Number of Companies Worked provides insights into how employees' previous work experience correlates with attrition rates. Employees with experience in **one company prior to the current organization** show the highest attrition rate, comprising **36.05% of all attrition** in the company. This suggests that such employees, likely in early stages of their careers, may be more open to exploring new opportunities. On the other hand, employees with experience in **five or more companies** also exhibit high attrition within their respective groups.

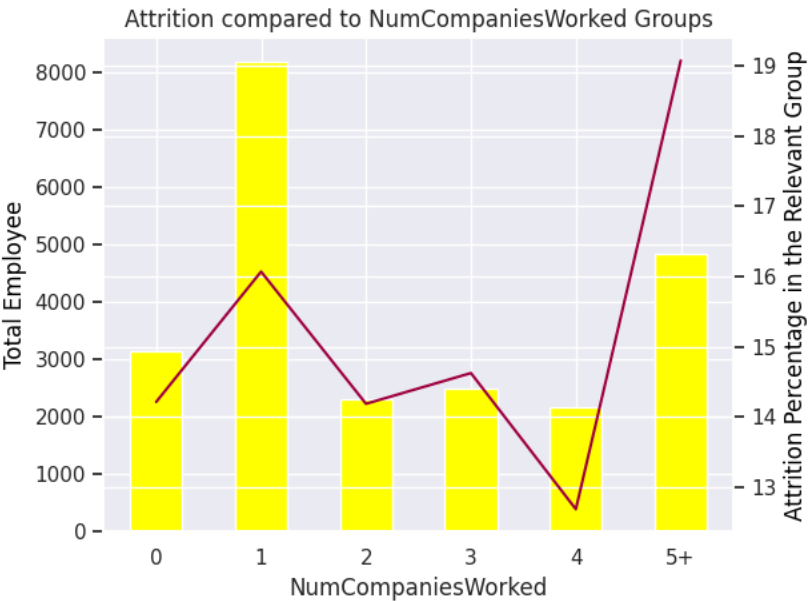


Fig 26: Graph showing attrition compared to number of companies worked

Total Working Years

✓ [152] # Total Working Years & Attrition Table
table_df(x = 'TotalWorkingYears')

	TotalWorkingYears	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	1	1465	473	32.286689	12.948262
1	2-5	3502	637	18.189606	17.437722
2	6-10	9585	1456	15.190402	39.857651
3	11-15	2978	374	12.558764	10.238160
4	16-20	2365	328	13.868922	8.978921
5	20+	3244	385	11.868064	10.539283

Fig 27: Attrition compared to total working hours

The "Total Working Years" feature captures the professional experience of employees in terms of years. Analysis of this feature reveals a significant pattern in attrition rates. Employees with one year or less of working experience exhibit the highest likelihood of leaving the company, accounting for 12.94% of total attrition. This could be indicative of their early career phase, during which employees often explore various opportunities and industries to find a suitable fit for their skills and aspirations.

Additionally, employees with six to ten years of experience represent the second-highest attrition group, comprising 39.8% of the total attrition across the company. This trend may reflect mid-career employees seeking growth opportunities, better compensation, or more fulfilling roles elsewhere.

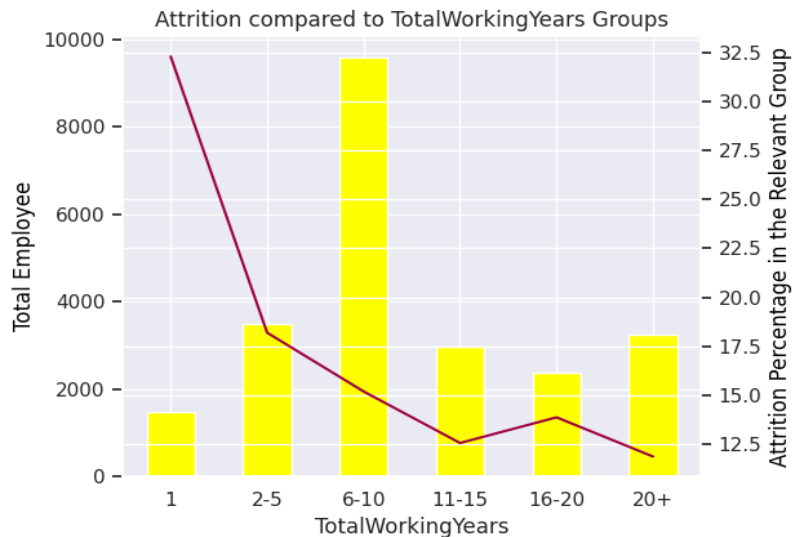


Fig 28: Graph showing attrition compared to total working hours

Years at Company

```
[160] # Years at Company & Attrition Table
table_df(x = 'YearsAtCompany')
```

	YearsAtCompany	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	1	3408	813	23.855634	22.255680
1	2-5	8799	1389	15.785885	38.023542
2	6-10	7045	975	13.839603	26.690391
3	10+	3887	476	12.245948	13.030386

Fig 29: Attrition compared to years at company

The "Years at Company" feature provides insights into the duration employees have spent working in the organization. Employees with one year or less of experience at the company exhibit the highest attrition percentage within their group, standing at 23.8%. This suggests a significant challenge in retaining employees during their initial year. Furthermore, employees with 2-5 years of experience at the company account for the highest number of attrition cases, with 1,389 employees leaving, making up 38.02% of all attrition across the organization. These findings highlight the need to address employee retention strategies, particularly during the early years of employment.

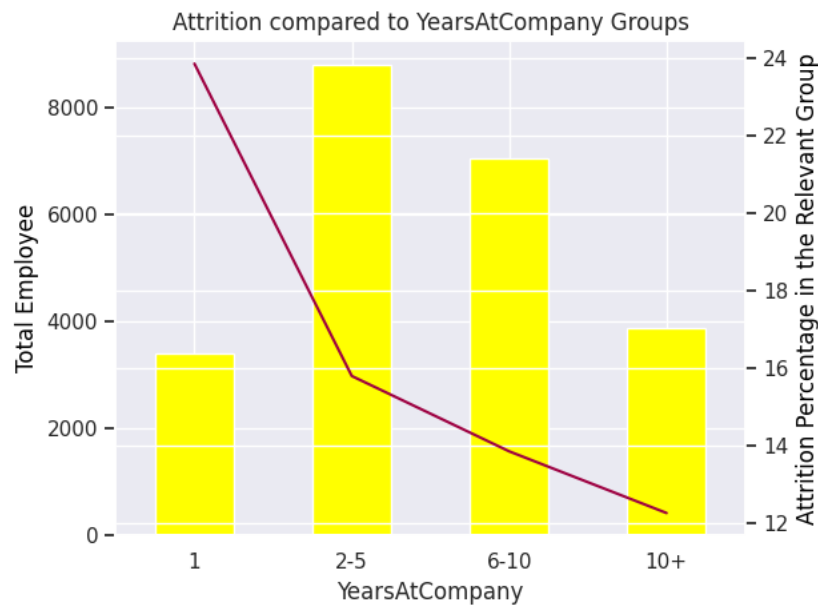


Fig 30: Graph showing attrition compared to years at company

Years Since Last Promotion

```
[166] # Years Since Last Promotion & Attrition Table
table_df(x = 'YearsSinceLastPromotion')
```

	YearsSinceLastPromotion	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	0	9159	1554	16.966918	42.540378
1	1	5612	841	14.985745	23.022174
2	2	2509	409	16.301315	11.196277
3	3	831	123	14.801444	3.367096
4	4-6	2155	271	12.575406	7.418560
5	7	1204	217	18.023256	5.940323
6	8+	1669	238	14.260036	6.515193

Fig 31: Attrition compared to years since last promotion

The "Years Since Last Promotion" feature reveals interesting attrition patterns within the company. Employees who have been with the company for less than one year since their last promotion are the most likely to leave, accounting for 42.5% of total attrition. This could be due to dissatisfaction or unmet expectations following their promotion. Employees with one and two years since their last promotion also exhibit high attrition numbers, reflecting a combined tendency for dissatisfaction or readiness to seek better opportunities. Notably, employees with seven years since their last promotion display the highest attrition rate within their group, suggesting a potential stagnation effect for those

who have remained unpromoted for an extended period. These insights highlight the need for targeted interventions to address dissatisfaction and stagnation across different tenure groups post-promotion.

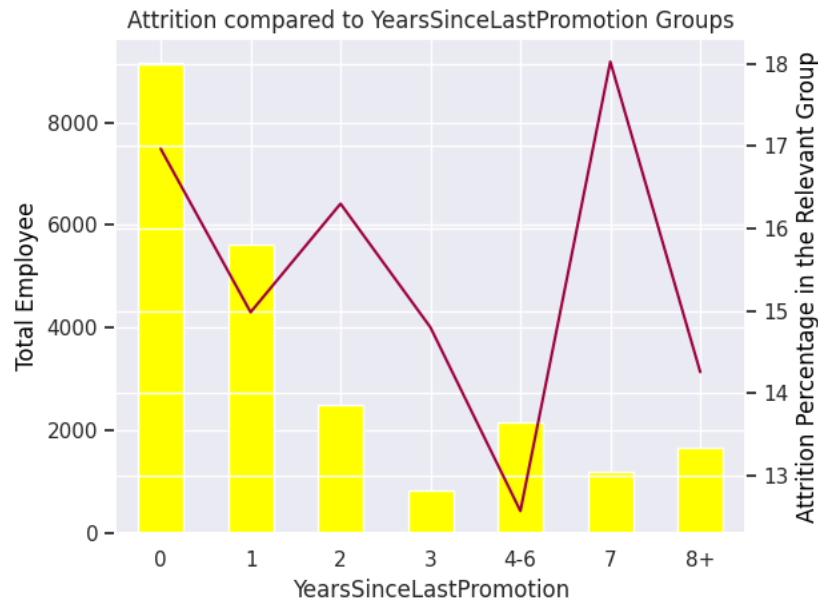


Fig 32: Graph showing attrition compared to years since last promotion

Feature/Variable Relationships

To understand the relationships between features in the dataset, a correlation matrix (visualized as a heatmap) was used. Correlation is a statistical measure that illustrates how strongly two variables are related. It ranges from -1 to +1, where a positive value indicates that as one variable increases, the other also increases, while a negative value indicates an inverse relationship. A value close to zero suggests no correlation.

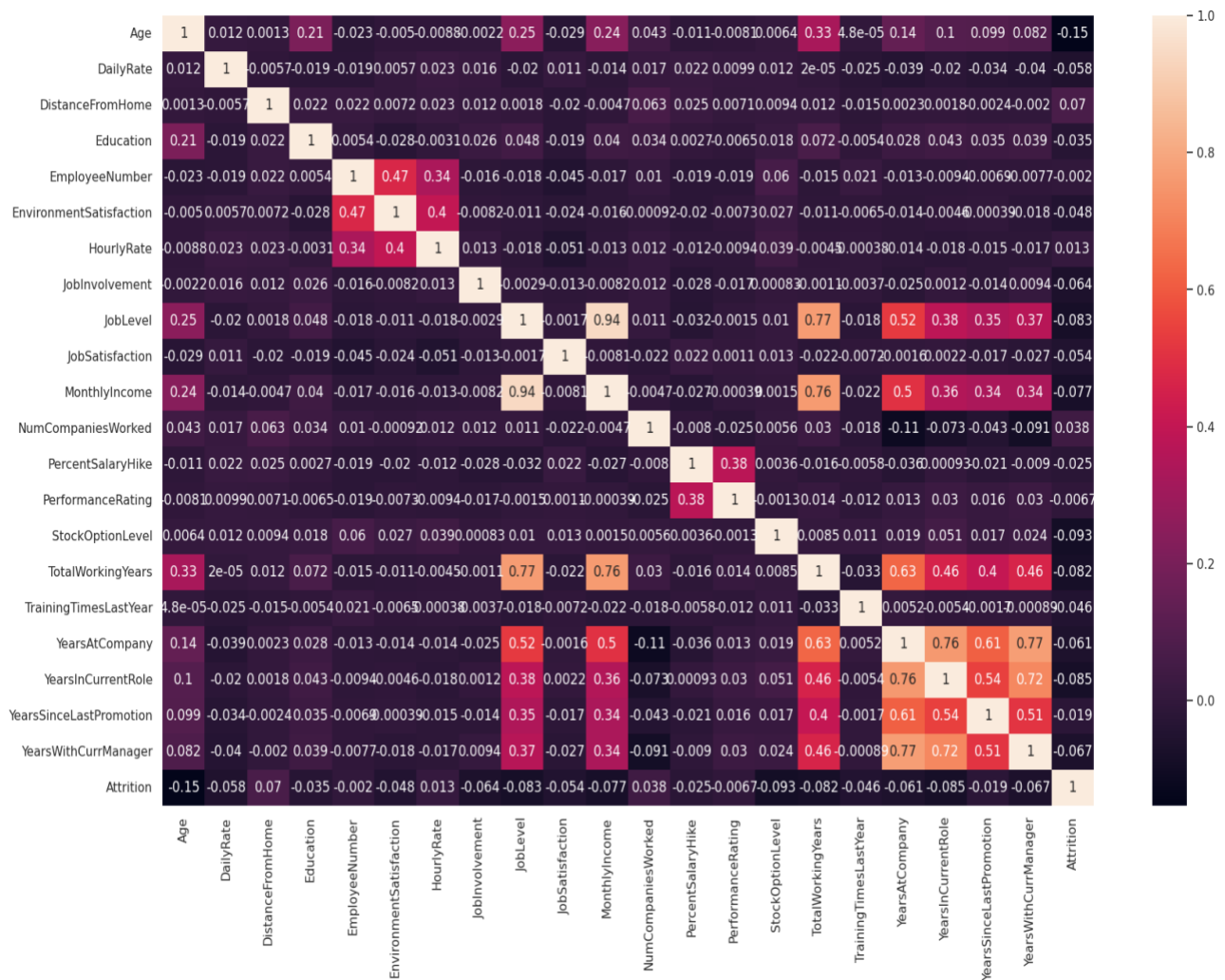


Fig 33: Correlation heat map showing variable relationship

The analysis revealed the following key findings:

Strong Correlations

- **Percent Salary Hike and Performance Rating:** A significant positive correlation exists between these two variables, suggesting that higher salary hikes are associated with higher performance ratings.
- **Total Working Years, Monthly Income, and Job Level:** These variables are strongly interrelated, indicating that employees with more years of experience tend to have higher job levels and income.
- **Years at Company, Years in Current Role, and Years with Current Manager:** These variables show strong interconnections, reflecting the natural overlap between tenure-related features.

Moderate Correlations

- **Age:** Moderately correlated with Total Working Years, Monthly Income, and Job Level, showing that older employees tend to have more experience and higher positions.
- **Job Level:** Moderately associated with Years at Company and Age, highlighting its relationship with seniority and tenure.
- **Total Working Years:** This feature is moderately linked with Years with Current Manager, Years Since Last Promotion, and other tenure-related variables.
- **Years at Company:** Displays moderate correlations with Years Since Last Promotion, Monthly Income, and Job Level, emphasizing its connection to career progression.
- **Years Since Last Promotion:** Shares moderate relationships with features like Years with Current Manager and Years in Current Role, suggesting a connection to career growth dynamics.

While examining correlations is essential for feature selection in predictive modelling, caution must be exercised to avoid multicollinearity issues. Strongly correlated variables, such as Years at Company, Years in Current Role, Years Since Last Promotion, and Years with Current Manager, may introduce redundancy and inflate variance in the model. It is advisable to evaluate these features individually and select those that enhance model performance.

The heatmap serves as a valuable tool for identifying these relationships visually. It provides insights into patterns that can guide feature engineering and model optimization. For example, age and experience-related variables often cluster together, reflecting their inherent dependency. By leveraging these insights, future steps can involve refining feature selection and mitigating multicollinearity, ensuring a robust predictive model.

Statistical Validation Techniques

HYPOTHESIS TESTING

Hypothesis testing is a fundamental statistical technique used to assess whether a certain assumption about a population parameter holds true. In this project, hypothesis testing is applied to evaluate whether there is a statistically significant difference in attrition rates between male and female employees. This technique is essential for providing evidence-based conclusions and for determining whether observed differences in data are due to random chance or underlying patterns.

Selecting the Appropriate Test

We use a **two-proportion z-test** to compare the attrition rates between male and female employees. This test is appropriate for evaluating the difference in proportions between two groups, especially when the sample size is large, as it ensures the sampling distribution approximates normality due to the Central Limit Theorem.

The Null and Alternate Hypotheses

The hypotheses are defined as follows:

- **Null Hypothesis (H_0):** There is no significant difference in the attrition rates for male and female employees.

$$H_0: p_{\text{male attrition}} - p_{\text{female attrition}} = 0$$

- **Alternative Hypothesis (H_a):** There is a significant difference in the attrition rates for male and female employees.

$$H_a: p_{\text{male attrition}} - p_{\text{female attrition}} \neq 0$$

The confidence level is set at 95%, corresponding to a significance level (α) of 0.05.

Computing the Test Statistic (z-score)

The z-score indicates how many standard deviations the observed difference in proportions is from the hypothesized difference under the null hypothesis. The formula is:

$$z = \frac{\text{Difference in Means}}{\sqrt{p(1-p)\left(\frac{1}{n_{\text{male}}} + \frac{1}{n_{\text{female}}}\right)}}$$

Calculating the p-value

The p-value represents the probability of obtaining a z-score as extreme as the observed one under the null hypothesis. For a two-tailed test:

$$p = 2 * P(Z > |z|) = 0.095$$

Margin of Error and Confidence Interval

The margin of error quantifies the uncertainty around the observed difference in proportions:

$$\text{Margin of Error} = 1.96 * 0.0048 = 0.0096$$

$$\text{Confidence Interval} = \text{Difference in Means} \pm \text{Margin of Error}$$

Computed Interval: [-0.0014, 0.0178]

The p-value (0.095) exceeds the significance level ($\alpha=0.05$), meaning we fail to reject the null hypothesis. Thus, there is insufficient evidence to conclude that male and female attrition rates differ significantly.

Machine Learning Techniques

Machine learning techniques form the cornerstone of predictive modelling for employee attrition. However, their success relies heavily on the quality of the data and the preparation processes that precede modelling. The pre-modelling phase is a critical step where the dataset is refined, transformed, and optimized to ensure the machine learning algorithms can effectively uncover meaningful patterns. Key tasks in this phase include feature engineering and selection, which aim to enhance the relevance and usability of the dataset.

Feature Engineering and Selection

Feature engineering and selection are pivotal in structuring the dataset to improve model performance. These processes focus on identifying and transforming variables that hold predictive potential while removing or refining features that may introduce noise or redundancy.

Deciding What Features to Keep

The dataset initially contained numerous features, not all of which were relevant or necessary for the model. Identifying the most impactful variables involved both domain knowledge and exploratory data analysis. Features such as Job Satisfaction, Work-Life Balance, and Monthly Income were

retained due to their direct correlation with employee behavior and potential attrition. Similarly, variables like Years at Company, Years in Current Role, and Years Since Last Promotion were included for their relevance to tenure and career progression.

Some features, however, were deemed irrelevant and excluded. For instance, EmployeeNumber served only as a unique identifier and did not contribute to predicting attrition. Additionally, highly correlated features were flagged during exploration for potential removal later, to address multicollinearity and improve model efficiency. These decisions ensured that the final dataset was streamlined, retaining features most likely to impact attrition predictions.

Transforming Categorical Data into Numerical

Since machine learning models operate on numerical data, categorical variables in the dataset were transformed to facilitate their use in the models. This transformation preserved the information within these features while converting them into a format suitable for computation.

Categorical variables such as BusinessTravel, Department, and EducationField were converted into numerical representations using one-hot encoding. This technique created separate binary columns for each category within a variable, ensuring the model treated them as independent groups without imposing a false hierarchy. For instance, BusinessTravel, with categories like "Travel_Rarely" and "Travel_Frequently," was split into multiple binary columns, each indicating the presence or absence of a specific category. For ordinal variables, such as Education, which naturally follow an order ("Below College" to "Doctorate"), label encoding was applied. This method assigned numeric values to the categories while maintaining their inherent rank. Transforming these variables was crucial for ensuring their compatibility with machine learning models and preserving the integrity of the relationships they represent.

By focusing on relevant features and accurately transforming categorical data, the pre-modeling process created a dataset primed for effective analysis. These steps laid a strong foundation for the subsequent phases, enabling the machine learning algorithms to deliver accurate and meaningful predictions.

Data Preprocessing

Splitting the Dataset into Training and Test Sets

Splitting the dataset is a foundational step in machine learning to evaluate model performance effectively. The dataset was divided into a training set (75%) and a test set (25%) using `train_test_split` from Scikit-learn. This ensures that the model learns from one part of the data and is evaluated on unseen data, reducing the chances of overfitting. A random state was set for consistency in results across different runs.

Feature Scaling

Feature scaling is crucial when dealing with features that have different ranges, ensuring that algorithms like SVM, K-NN, and PCA perform optimally. The `StandardScaler` from Scikit-learn was used, which standardizes features by removing the mean and scaling them to unit variance. Both the training and test sets were scaled, with the scaler fitted on the training data and applied to the test set for consistency.

Handling Imbalanced Datasets

In the dataset, there was a significant imbalance in the target variable, with 84% of instances labeled as "No" (0) and only 16% as "Yes" (1). To address this, the SMOTE (Synthetic Minority Oversampling Technique) method was applied. SMOTE synthesizes new samples for the minority class, balancing the dataset without losing information. This step ensures the model does not favor the majority class excessively and improves its ability to predict the minority class.

Principal Component Analysis (PCA)

PCA was employed to reduce the dimensionality of the dataset while retaining most of the variance. All features were transformed into principal components, and the explained variance ratio for each component was calculated. The cumulative variance showed that 27 components were required to explain over 90% of the variation in the data. Although no natural cutoff point for dimensionality reduction was identified, this step optimized the feature space and reduced redundancy, enhancing computational efficiency for downstream models. The components were visualized to provide insights into their variance contribution.

This preprocessing pipeline ensured that the data was clean, balanced, and ready for effective model training and evaluation. Each step was implemented with the aim of improving the robustness and reliability of the machine learning models used in the subsequent analysis.

Applying Machine Learning Algorithms

To predict employee attrition effectively, various machine learning algorithms were implemented and evaluated. Each algorithm brings unique strengths to the table, making it essential to test their performance in the context of this project. The evaluation emphasized the ability of each model to handle imbalanced datasets and accurately predict the minority class, which in this case represents employee attrition.

Logistic Regression

Logistic Regression, a widely used statistical model, predicts binary outcomes by modelling the probability of a given class. It assumes a linear relationship between the independent variables and the log-odds of the dependent variable. In this project, Logistic Regression served as a baseline model due to its simplicity and interpretability. Both the full feature set and the Principal Component Analysis (PCA)-reduced features were used during training. Hyperparameter tuning focused on the regularization parameter, optimizing the model's ability to generalize without overfitting.

Decision Tree Classification

Decision Tree Classification is a versatile algorithm that splits the dataset into subsets based on feature values, forming a tree structure. It identifies the most critical features contributing to predictions, making it highly interpretable. Decision trees were trained on both the full feature set and PCA-transformed features. Hyperparameters like tree depth and the minimum number of samples per split were optimized to prevent overfitting and improve accuracy. This model provided insights into the hierarchy of feature importance in predicting attrition.

Random Forest Classification

Random Forest, an ensemble learning method, constructs multiple decision trees during training and aggregates their outputs for predictions. By leveraging the power of multiple trees, it reduces the risk of overfitting and enhances prediction accuracy. In this project, Random Forest was trained on all features and PCA-transformed features. Hyperparameters such as the number of trees (`n_estimators`) and maximum tree depth were tuned. This model demonstrated robustness and was among the top performers in terms of F1 score.

K-Nearest Neighbours (K-NN) Classification

K-NN is a non-parametric algorithm that classifies a data point based on the majority class among its k-nearest neighbours. It is simple yet effective for datasets where relationships between data points are crucial. In this project, K-NN was applied to both the full and PCA-reduced datasets. Hyperparameter tuning involved selecting the optimal number of neighbours (k) and evaluating the model's sensitivity to feature scaling. K-NN performed moderately well, highlighting its dependence on the choice of k and the quality of scaled data.

Support Vector Machine (SVM) Classification

Support Vector Machines aim to find the hyperplane that best separates classes with maximum margin. This algorithm is particularly effective for datasets with clear boundaries between classes. Both linear and kernel-based SVMs were implemented in this project. Kernel SVM, with a radial basis function (RBF) kernel, captured non-linear relationships in the data. Hyperparameter tuning focused on the regularization parameter and the kernel coefficient (gamma). SVM emerged as one of the best-performing models, achieving a high F1 score.

Naive Bayes Classification

Naive Bayes is a probabilistic model based on Bayes' theorem, assuming independence among predictors. Despite its simplicity and the unrealistic assumption of feature independence, Naive Bayes is computationally efficient and effective for certain datasets. In this project, Naive Bayes was tested with both the original and PCA-reduced features. While it provided quick results, its performance was slightly lower than other algorithms, emphasizing the influence of correlated features on its predictions.

Gradient Boosting Classification

Gradient Boosting is an advanced ensemble technique that sequentially builds models, with each iteration correcting the errors of its predecessor. This approach allows the algorithm to focus on difficult-to-predict samples, improving overall accuracy. Gradient Boosting was trained on both full and PCA-reduced datasets. Hyperparameters such as the learning rate, the number of estimators, and maximum tree depth were optimized to balance complexity and accuracy. This model showed consistent and robust performance.

ADA Boost Classification

ADA Boost is another boosting technique that emphasizes misclassified samples during training, improving the model's focus on challenging cases. In this project, ADA Boost was applied to both feature sets, with hyperparameters like the learning rate and number of estimators fine-tuned for optimal performance. While its results were competitive, the algorithm's sensitivity to noise in the data occasionally affected its performance.

Performance Evaluation Metrics

In machine learning, evaluating model performance is crucial, particularly when dealing with imbalanced datasets where one class significantly outweighs the other. A dataset with an unequal distribution of classes poses unique challenges, as traditional evaluation metrics like accuracy can provide misleading results. For example, a model predicting only the majority class could achieve high accuracy while failing to identify the minority class entirely. In this project, where predicting employee attrition is the primary goal, the minority class (employees who left the company) holds critical importance. Hence, more nuanced metrics such as precision, recall, and F1 score were used to assess the models effectively.

Precision, Recall, and F1 Score

Precision and recall are foundational metrics in classification problems. Precision measures the proportion of correctly predicted positives (employees predicted to leave) out of all predicted positives, providing insights into how well the model avoids false positives. Recall, on the other hand, measures the proportion of correctly identified positives out of all actual positives, focusing on minimizing false negatives. These two metrics often conflict, as improving one may negatively affect the other.

To strike a balance between precision and recall, the **F1 score** was selected as the primary evaluation metric. The F1 score is the harmonic mean of precision and recall, ensuring that both metrics are weighted equally. It is particularly valuable in imbalanced datasets, as it emphasizes the accurate prediction of the minority class without neglecting the overall performance. The formula for the F1 score is:

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Evaluation Process

Each model implemented in this project was assessed using the F1 score to ensure its ability to predict employee attrition effectively. Confusion matrices were also employed to provide a detailed view of model performance by illustrating the counts of true positives, true negatives, false positives, and false negatives. These matrices helped identify specific areas where the models struggled, such as incorrectly predicting employees who did not leave as those who did (false positives) or missing actual cases of attrition (false negatives).

The dataset's inherent imbalance was addressed using techniques like Synthetic Minority Oversampling Technique (SMOTE) during preprocessing, which complemented the evaluation process by ensuring that the models were trained on a balanced dataset. This adjustment reduced the impact of the class imbalance on the F1 score and improved the reliability of the evaluations.

Ethical Considerations

Ethical considerations are vital when developing machine learning models, particularly in sensitive areas like human resources. This project prioritized the privacy and confidentiality of employee data by ensuring that no personally identifiable information (PII) was used and by adhering to data protection regulations such as GDPR. Measures were taken to prevent bias, ensuring that features like gender and marital status did not unfairly influence predictions, thereby fostering fairness and inclusivity.

Transparency and accountability were also integral to the process. Efforts were made to use interpretable models where possible, while advanced methods like feature importance plots were employed for complex models to provide clarity on decision-making. The model's predictions were designed to support proactive HR interventions, such as identifying employees at risk of attrition and offering support, rather than punitive measures. By emphasizing fairness, privacy, and accountability, the project aims to deliver actionable insights responsibly, ensuring alignment with ethical principles and organizational values.

RESULTS AND DISCUSSION

This chapter provides a detailed analysis of the results obtained from the implementation of machine learning models for predicting employee attrition. It also discusses the model comparison, feature importance, and their implications in the context of the dataset.

Model Comparison

Nine machine learning algorithms were employed to predict attrition, including Logistic Regression, Decision Tree, Random Forest, K-NN, SVM, Kernel SVM, Naive Bayes, Gradient Boosting, and AdaBoost. Each model was evaluated using four configurations:

- All Features
- Tuned All Features
- PCA Features
- Tuned PCA Features.

	all_features	tuned_all_features	PCA_features	tuned_PCA_features	Total Improvement(%)
Logistic Regression	0.85218	0.84391	0.86737	0.85761	0.0178
Decision Tree	0.76488	0.77981	0.76138	0.79637	0.0412
Random Forest	0.82035	0.83650	0.82247	0.83080	0.0197
K-NN	0.80187	0.78739	0.81878	0.81101	0.0211
SVM	0.87818	0.87684	0.86794	0.87766	0.0000
Kernel SVM	0.80354	0.80784	0.80181	0.81301	0.0118
Naive Bayes	0.71555	0.71085	0.73315	0.71427	0.0246
Gradient Boosting	0.84124	0.85171	0.83564	0.86209	0.0248
ADA Boosting	0.82298	0.85948	0.85089	0.82795	0.0444

Fig 34: Comparing F1-scores of models

The models were assessed using the F1-score, emphasizing the ability to accurately identify the minority class (employees leaving the organization). Among the models, the tuned **Support Vector Machine (SVM)** with a linear kernel emerged as the best performer, achieving the highest F1-score of 0.88. This superior performance highlighted the model's ability to handle imbalanced datasets effectively. Other models, including **Random Forest** and **Gradient Boosting**, also performed well, reinforcing the importance of ensemble techniques in predictive analytics.

The comparison demonstrated that hyperparameter tuning significantly improved model performance, emphasizing the necessity of iterative optimization for achieving robust results.

Ensemble models, such as Gradient Boosting and AdaBoost, showcased their strength in handling complex patterns in the dataset, albeit with slightly lower scores compared to SVM.

Feature Importance

A critical aspect of this project was understanding the factors contributing to employee attrition. Feature importance analysis, based on coefficients derived from the SVM model, provided insights into the most significant predictors of attrition.

```
➡ {'Age': -1.4746034447649103e-05,  
  'BusinessTravel': 2.0601761914679217e-05,  
  'DailyRate': 4.611157128268234e-05,  
  'Department': -1.2010143531637762e-06,  
  'DistanceFromHome': -3.405620315477531e-06,  
  'Education': 7.494825538190142e-06,  
  'EducationField': 1.8022937581602116e-05,  
  'EnvironmentSatisfaction': -0.00010576040184719082,  
  'Gender': 1.0383074789288571e-05,  
  'HourlyRate': 7.930734174182996e-05,  
  'JobInvolvement': -1.7103946593988173e-05,  
  'JobLevel': 2.9811332159943027e-05,  
  'JobRole': -2.1755289172767367e-05,  
  'JobSatisfaction': -1.945536009457377e-05,  
  'MaritalStatus': 1.518226153129909e-06,  
  'MonthlyIncome': -2.767464240605033e-05,  
  'MonthlyRate': -7.332459138975622e-06,  
  'NumCompaniesWorked': 7.130445545011632e-06,  
  'OverTime': 3.337379892857234e-05,  
  'PercentSalaryHike': -1.9023548915524102e-05,  
  'RelationshipSatisfaction': 6.773579275226593e-06,  
  'StockOptionLevel': 1.571222844040654e-05,  
  'TotalWorkingYears': 2.3643917374444046e-05,  
  'TrainingTimesLastYear': 1.8300583506325463e-05,  
  'WorkLifeBalance': 9.038893388230163e-05,  
  'YearsAtCompany': 9.680158105224157e-05,  
  'YearsInCurrentRole': -0.00010590065730942122,  
  'YearsSinceLastPromotion': -6.13739073935804e-05,  
  'YearsWithCurrManager': -7.258782444724554e-05}
```

Fig 35: Output data showing feature importance with keys and values

The analysis revealed that **Environment Satisfaction** was the most influential factor, exhibiting a strong negative correlation with attrition. Employees reporting higher satisfaction with their work environment were less likely to leave, highlighting the importance of fostering a positive workplace culture. Similarly, **Years in Current Role** demonstrated a significant negative relationship with attrition. Employees with longer tenure in their current roles showed greater organizational commitment, reducing their likelihood of leaving.

Years Since Last Promotion emerged as another key factor, with a longer duration since the last promotion correlating with higher attrition rates. This finding underscores the role of career

progression in employee retention. **Monthly Income** also played a crucial role, with higher incomes generally associated with reduced attrition. However, specific income ranges displayed anomalies, indicating that compensation, while important, must be complemented by other retention strategies.

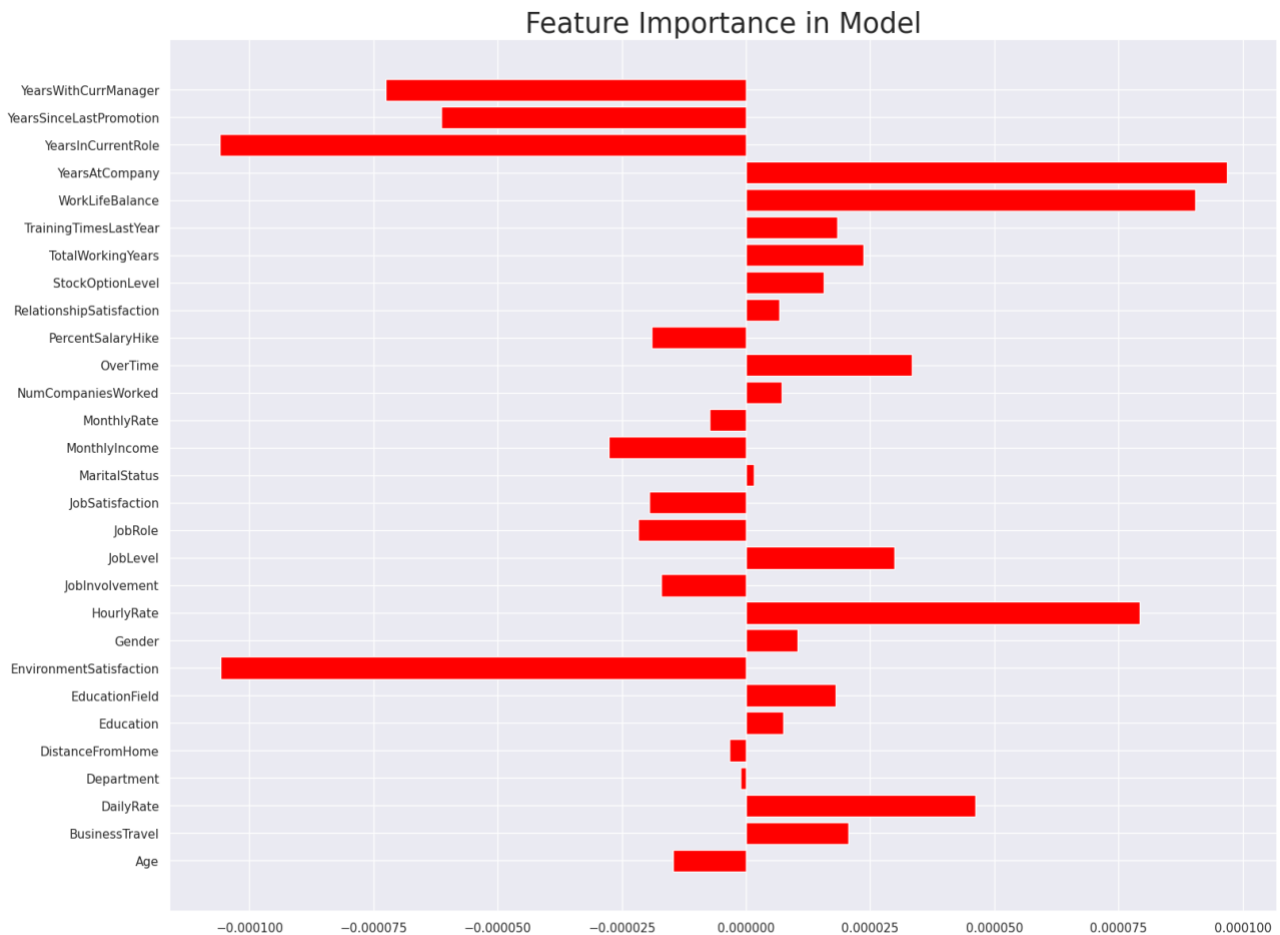


Fig 36: Graph showing feature importance

Work-Life Balance was another critical factor, as employees struggling to balance their professional and personal lives were more likely to leave. Other notable contributors included **Over Time** and **Training Times Last Year**, where excessive overtime and inadequate training were linked to higher attrition.

A feature importance graph visually illustrated these findings, emphasizing the relative impact of each variable. Key drivers like Environment Satisfaction, Years in Current Role, and Work-Life Balance stood out as areas requiring organizational attention to reduce attrition effectively.

Discussion

The findings from this chapter provide actionable insights for organizations aiming to improve employee retention. Models like SVM and Random Forest demonstrated exceptional predictive capabilities, while feature importance analysis highlighted critical areas for intervention. By focusing on improving job satisfaction, ensuring timely promotions, maintaining work-life balance, and providing competitive compensation, organizations can address the root causes of attrition.

Furthermore, the results underscore the importance of integrating machine learning techniques with feature analysis to derive meaningful insights. Predictive models not only identify at-risk employees but also guide strategic decisions that enhance organizational performance and employee satisfaction. This approach bridges the gap between analytics and actionable HR strategies, ensuring data-driven decision-making aligns with organizational goals.

Recommendations to Improve the Model and Future Scope

Recommendations to Improve the Model

Enhancing the model's predictive accuracy requires improvements in data collection and processing. The company should prioritize gathering additional data through targeted surveys. Exit surveys can provide insights into why employees leave, while surveys targeting long-tenured employees can highlight factors promoting retention. Regular surveys, such as monthly or biannual assessments, can capture evolving patterns and trends, ensuring the dataset remains current and comprehensive.

Preprocessing techniques, such as advanced feature engineering and hyperparameter tuning, can refine the model further. Addressing class imbalance through effective resampling methods will also improve the model's ability to predict minority classes accurately. Increasing the quantity of observations is critical, as larger datasets enable the model to generalize better. Incorporating external data sources, such as industry benchmarks, can provide additional context to enrich the dataset. Additionally, adopting explainable AI (XAI) techniques will enhance transparency, enabling stakeholders to understand the factors driving predictions.

Future Scope

This project holds significant potential for future development. Integrating the model into real-time attrition monitoring systems can provide dynamic insights, allowing the organization to take timely action. The findings can guide proactive HR strategies, such as personalized retention programs targeting at-risk employees, addressing concerns like work-life balance and career growth.

Future iterations of the project could explore advanced machine learning techniques, including deep learning and ensemble methods, to improve accuracy and capture complex feature interactions. Integrating the model into HR systems would automate the attrition prediction process, making data-driven insights accessible for strategic decision-making. The methodologies applied in this project can also extend to other HR functions, such as recruitment and performance management, fostering a comprehensive, data-driven approach across the organization.

CONCLUSION

This study explored the potential of machine learning techniques to address employee attrition by analyzing HR data and predicting factors influencing turnover. By leveraging a robust dataset and advanced analytical methods, the research identified key drivers of attrition and developed effective predictive models to mitigate workforce instability.

The findings emphasized the importance of factors such as years since last promotion, work-life balance, and job satisfaction in determining attrition risk. Models like SVM, Random Forest and Gradient Boosting demonstrated superior performance, achieving high accuracy and F1 scores, underscoring their reliability for predicting employee turnover. Through feature importance analysis, actionable insights were derived, enabling HR professionals to implement targeted interventions to improve retention.

Moreover, the study highlighted the critical role of data preprocessing, feature selection, and evaluation metrics in ensuring the robustness of predictive models. Ethical considerations were central to the research, ensuring the responsible use of employee data while complying with privacy regulations. Recommendations were provided for collecting more granular and diverse data through structured surveys and regular feedback mechanisms, which could enhance model precision and organizational decision-making.

The study also acknowledged limitations, including the reliance on synthetic datasets and the challenges of generalizing results to real-world scenarios. Future research can address these gaps by incorporating external economic factors, behavioural data, and longitudinal analyses to improve prediction accuracy and applicability. Advancements in machine learning, such as explainable AI, could further enhance transparency and trust in predictive models.

In conclusion, this research demonstrates the transformative potential of HR analytics powered by machine learning in predicting and addressing employee attrition. By adopting data-driven strategies, organizations can foster a proactive approach to workforce management, ensuring long-term stability, enhanced employee satisfaction, and sustained business success. The integration of predictive analytics into HR practices marks a significant step forward in creating resilient and agile workplaces.

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APPENDICES

Appendix A: Research Project Plan

Task 2: Research Proposal

Applying Machine Learning for Employee Retention: Predictive Analytics, HR Strategies, and Future Approaches to Mitigating Attrition

Introduction and Justification:

Employee retention is essential for organisational success, since it directly impacts productivity, operational efficiency, and financial performance (Lawrencia et al., 2024). Although this is crucial, Amarakoon and Colley (2022) argued that acquiring and keeping highly qualified employees remains one of the toughest challenges for organisations around the world in the twenty-first century. In this regard, companies must develop efficient strategies and tools to accurately identify employees who are at risk of leaving and address their individual needs (Gazi et al., 2024). Several factors influence retention, including job embeddedness (Noordin et al., 2021), working conditions (Mosadeghrad, 2013), and socially responsible leadership (Arishi et al., 2018). Effective retention strategies are thus critical for enhancing employee-employer relationships and increasing organisational profitability (Kuume & Angula, 2020). Organisations that apply these complete retention efforts can minimise turnover while also increasing profitability (Chauhan, 2024). Furthermore, prioritising job happiness, employee engagement, and organisational commitment can greatly boost retention efforts (Irabor & Okolie, 2019). In this context, machine learning can be an effective tool for improving retention methods. Machine learning can create predictive models that identify employees who are at risk of attrition, allowing businesses to take proactive efforts to address possible issues before they lead to turnover (Musunga, 2022).

Research Aim:

This research aims to investigate the application of machine learning in predicting loss of employees, identifying critical factors affecting staff turnover, and evaluating the efficacy of

predictive models in practical HR environments. The study will offer actionable recommendations for HR professionals to formulate targeted retention strategies and enhance workforce stability.

Research Question:

How can machine learning be utilised to foresee employee attrition, and what actionable insights can HR professionals derive to enhance staff retention effectively?

Research Objectives:

1. To identify key factors contributing to employee attrition by analysing demographic data, workspace conditions, job characteristics and performance related metrics.
2. To build and validate machine learning models that accurately and reliably predict employee turnover.
3. To evaluate the ethical and privacy concerns associated with employing predictive analytics in HR decision-making.
4. To offer practical strategies for HR teams to effectively leverage predictive insights to improve employee retention and cultivate a supportive workplace culture.
5. To explore future trends and opportunities for the application of machine learning in HR, emphasising the enhancement of workforce stability and satisfaction.

Deliverables:

This research will deliver a thorough investigation that offers evidence-based insights into the utilisation of machine learning for predicting and mitigating employee attrition. The findings will encompass:

1. A meticulous analysis of the principal factors contributing to employee turnover and their identification through data-driven methodologies.
2. The development and assessment of predictive models to accurately forecast employee attrition.
3. Strategic recommendations for HR professionals to utilise these insights in executing effective retention programs.
4. A framework for incorporating machine learning tools within HR practices to improve decision-making and workforce planning.

Future directions for enhancing predictive analytics in HR while addressing challenges associated with data quality, ethics, and employee privacy.

Literature Review

Introduction

Organisations have long been concerned about employee attrition, or turnover, because it has a big influence on financial performance, morale, and productivity. Conventional approaches to attrition management frequently depended on surveys and retrospective assessments, which offered little understanding of the nuances of employee behaviour. However, HR professionals can now use sophisticated predictive models to identify attrition risks and create successful interventions thanks to the development of machine learning (ML) and big data analytics. In addition to discussing important contributing elements and ethical issues with using employee data for predictive analytics, this paper examines current developments in machine learning approaches for forecasting employee attrition.

Machine Learning Models for Predicting Attrition

Recent research highlights how well different machine learning algorithms predict employee turnover. The use of well-known techniques including support vector machines (SVM), decision trees, random forests, and neural networks is highlighted by Al-Alawi and Ghanem (2024). Because Random Forest and XGBoost are ensemble learning techniques that can handle unbalanced datasets and capture intricate feature interactions, they routinely beat conventional algorithms. Artificial neural networks (ANNs) and recurrent neural networks (RNNs), two deep learning techniques, have also shown promise in identifying non-linear relationships in HR datasets, especially when working with unstructured data like employee feedback text (Alqahtani et al., 2024).

Support Vector machines work well in situations with well-defined decision boundaries and are especially useful for small-to-medium-sized datasets. Despite being straightforward and easy to understand, decision trees are frequently utilised as the foundation for more complex ensemble models. Al-Alawi and Ghanem (2024) contend that although individual models offer insightful information, ensemble techniques such as Random Forest and XGBoost perform better by combining predictions from several decision trees, which lessens overfitting and enhances generalisability.

Factors Influencing Employee Attrition

Numerous elements from the individual, organisational, and external domains have been found to have an impact on attrition. According to Alhamad et al. (2024), work-life balance, leadership efficacy, organisational culture, and job satisfaction are important indicators of employee turnover. Job satisfaction continues to be the most important factor among these, and disengaged or unhappy workers are more likely to quit their jobs. Attrition risks are further increased by organisational elements including inadequate leadership and a dearth of opportunity for advancement.

Work-life balance has become a crucial factor, especially in the post-pandemic era when flexible scheduling and hybrid work arrangements have grown commonplace. Businesses that fall short of these standards have increased employee turnover, particularly among younger workers (Alhamad et al., 2024). Furthermore, when the employment market is good, employees may look for better possibilities elsewhere, therefore outside variables like market circumstances and industry trends can have an impact on attrition.

Commonly Used datasets In Attrition Prediction

One of the most popular datasets for comparing machine learning models in attrition research is the IBM HR Analytics Employee Attrition dataset (Alqahtani et al., 2024). A wide range of elements are included in this dataset, such as employment roles, performance indicators, remuneration information, and demographic data. This dataset has been used by researchers to assess the performance of different machine learning algorithms, allowing for a consistent comparison of feature importance and predicted accuracy. The significance of feature engineering in improving model performance is emphasised by Alqahtani et al. (2024), who find that variables like "monthly income" and "job involvement" are powerful indicators of attrition.

Practical Implications of Predictive Analytics

Organisations can proactively develop targeted retention tactics by recognising early warning signals of turnover. By concentrating on high-risk workers who show early signs of disengagement, predictive models help HR departments better allocate resources. To address certain attrition reasons, for example, interventions like wellness programs, mentorship opportunities, and career development programs can be customised (Alhamad et al., 2024).

The concrete advantages of predictive analytics are illustrated through case studies from a range of businesses. Businesses in the technology sector have successfully lowered attrition rates by

incorporating insights from machine learning into their human resources procedures. For instance, HR teams may make data-driven decisions to improve employee engagement and happiness by using real-time dashboards driven by machine learning models.

Ethical Considerations In Employee Data Use

The utilisation of employee data raises ethical issues that should not be disregarded, despite machine learning's enormous potential in HR analytics. Dr. N.V. Balaji (2020) and Alhamad et al. (2024) stress the significance of upholding consent and transparency when gathering and evaluating employee data. When handling sensitive data pertaining to workers' personal and professional lives, organisations must make sure that data protection laws like the GDPR are followed.

Bias in predictive models is another critical concern. Unchecked biases in training data can lead to discriminatory outcomes, such as unfairly targeting specific demographic groups for retention efforts while neglecting others. Researchers advocate for rigorous bias detection and mitigation techniques, including fairness-aware ML algorithms and regular audits of model performance. Additionally, clear communication with employees about the purpose and benefits of predictive analytics can foster trust and mitigate resistance to data-driven HR practices.

Emerging Trends In HR Analytics

With new technology opening up fresh possibilities for attrition prediction, the discipline of HR analytics is still developing. Textual data from employee surveys and feedback is increasingly being analysed using natural language processing (NLP), which offers deeper insights into employee engagement and mood. In a similar vein, developments in explainable AI (XAI) are improving interpretability and confidence in machine learning systems by helping HR professionals comprehend the reasoning behind model predictions.

Alqahtani et al. (2024) also emphasise how ML models may be integrated with employee engagement platforms to create smooth processes for tracking and managing attrition concerns. Employee engagement and satisfaction have been demonstrated to increase with gamification of HR procedures backed by predictive analytics, especially in large companies with diverse workforces.

Conclusion

HR analytics has undergone a revolution thanks to machine learning, which provides effective solutions for anticipating and reducing staff churn. While deep learning techniques show promise for managing complicated and unstructured data, ensemble techniques like Random Forest and XGBoost lead the way in accuracy and dependability. Organisations can create retention strategies that meet employee requirements and organisational objectives by comprehending the elements that contribute to employee turnover and utilising predictive insights. To guarantee that predictive analytics procedures maintain openness, equity, and adherence to data protection laws, it is crucial to address ethical issues. One important step in building more sustainable and employee-focused workplaces is the ongoing incorporation of cutting-edge machine learning techniques into HR procedures.

Methodology

Overview

This research adopts a comprehensive methodology to explore employee attrition and its predictors using advanced machine learning techniques. By leveraging a publicly available dataset, the study investigates key factors contributing to attrition and develops predictive models to identify employees at risk of leaving. The methodology adheres to the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, ensuring a structured approach to data analysis and model development.

Research Design

The research employs a quantitative, explanatory design, focused on analyzing patterns and relationships between variables associated with employee turnover. The dataset, sourced from Kaggle, comprises 23,436 records and 37 features, encompassing demographic, job-related, and satisfaction indicators. The study's objectives include identifying significant predictors of attrition, developing machine learning models, and evaluating their performance.

The explanatory nature of the study ensures that relationships between features and attrition are not only identified but also quantified, enabling actionable insights. This design aligns with industry best practices, facilitating reproducibility and practical application, while allowing for meaningful generalization of the results across similar datasets.

Research Philosophy and Approach

The research is grounded in positivist philosophy, emphasizing objective data analysis and empirical validation. The study adopts a deductive approach, testing hypotheses derived from existing literature and exploratory data analysis (EDA). This approach ensures that the findings are both theoretically grounded and data-driven.

EDA plays a crucial role in uncovering patterns within the dataset, guiding feature engineering and model selection. By thoroughly analyzing the relationships and distributions within the data, the research identifies significant trends that support hypothesis formulation. The use of this dual approach enhances the study's robustness and ensures findings are supported by data-driven evidence.

Data Collection

The dataset used in this study, titled "IBM HR Analytics Employee Attrition & Performance," was retrieved from Kaggle. It provides a rich source of information on employee demographics, job roles, and satisfaction metrics. The initial dataset consisted of 23,436 rows and 37 features, but after preprocessing and cleaning, the data was reduced to 23,141 rows and 32 features, ensuring high data quality and relevance.

Key features include:

- **Demographic Information:** Variables like Age, Gender, and Marital Status provide insights into employee background characteristics.
- **Job-Related Metrics:** Attributes such as Job Role, Department, and Monthly Income are indicative of the employee's professional context.
- **Satisfaction Scores:** Measures like Environment Satisfaction, Job Satisfaction, and Relationship Satisfaction highlight workplace conditions and employee perceptions.
- **Tenure Information:** Variables such as Years at Company, Years in Current Role, and Years Since Last Promotion give insights into employee tenure and progression.

The dataset's balance of numerical and categorical features provides a comprehensive basis for machine learning analysis, enabling both interpretability and predictive accuracy.

Data Preprocessing

Preprocessing is a critical step to ensure the dataset is clean, consistent, and ready for analysis. The following techniques were meticulously applied:

- **Handling Missing Values**

The dataset underwent thorough inspection for missing values. Imputation methods were deemed unnecessary as the dataset was complete, with no missing information. This ensured the integrity of the data and minimized the risk of introducing bias during preprocessing.

- **Encoding Categorical Variables**

Categorical variables like Business Travel, Department, and Gender were transformed into numerical formats using techniques such as label encoding and one-hot encoding. This step was crucial for compatibility with machine learning algorithms, which require numerical inputs for efficient processing.

- **Addressing Class Imbalance**

The dataset exhibited class imbalance, with significantly fewer instances of employees leaving compared to those staying. To address this, the Synthetic Minority Oversampling Technique (SMOTE) was applied. SMOTE generates synthetic samples for the minority class, enhancing model performance by reducing bias toward the majority class.

- **Scaling and Normalization**

Feature scaling was applied using Min-Max scaling to normalize the range of numerical features. Variables like Monthly Income and Years at Company were scaled to ensure that features with larger ranges did not disproportionately influence the model training process.

Feature Engineering and Selection

Feature engineering and selection were integral to improving model interpretability and performance. Irrelevant or redundant features were removed, while meaningful transformations were applied to enhance predictive accuracy. The steps included:

- **Selecting Relevant Features:** A rigorous selection process identified variables that significantly impacted attrition prediction. Features with negligible correlations to the target variable were excluded.
- **Creating Derived Features:** New features were engineered to capture latent relationships within the data, such as ratios or interaction terms.
- **Evaluating Feature Importance:** Feature importance rankings, derived from tree-based models like Random Forest, guided the selection of high-impact variables such as Years Since Last Promotion and Job Satisfaction.

Machine Learning Modelling

Several machine learning algorithms were implemented to develop predictive models. Each algorithm was selected based on its suitability for classification tasks and its ability to handle imbalanced datasets. These models included:

- **Logistic Regression:** A simple yet effective baseline model providing insights into feature relationships and their influence on attrition.
- **Decision Tree Classification:** A non-parametric model known for its interpretability and ability to handle non-linear data relationships.
- **Random Forest Classification:** An ensemble method that improves accuracy by combining predictions from multiple decision trees.
- **K-Nearest Neighbours (K-NN):** A distance-based algorithm that predicts class membership based on the closest training examples.
- **Support Vector Machine (SVM):** Both linear and kernel SVMs were tested to classify complex patterns by maximizing the margin between classes.
- **Kernel SVM:** An extension of SVM using non-linear kernel functions to better capture intricate data patterns.
- **Naive Bayes Classification:** A probabilistic model based on Bayes' theorem, particularly effective for categorical data.
- **Gradient Boosting Classification:** An iterative ensemble method that builds models sequentially to correct errors from previous iterations.
- **ADA Boost Classification:** A boosting algorithm that adjusts the weights of misclassified samples, focusing on difficult-to-predict instances.

Each model underwent rigorous evaluation using metrics like accuracy, precision, recall, and F1-score. Random Forest, SVM and Gradient Boosting emerged as top performers, achieving high predictive accuracy and robustness against overfitting.

Performance Evaluation Metrics

To evaluate the classifiers' performance, the study prioritized F1-score as the primary metric. The F1-score balances precision and recall, making it ideal for imbalanced datasets. Additional metrics, such as the confusion matrix, provided complementary insights into model efficacy.

This evaluation strategy ensured a balanced assessment of the models' ability to correctly classify both minority and majority class instances.

Ethical Considerations

Ethical compliance was integral to this study. Adhering to data privacy regulations such as GDPR, the research ensured that sensitive information remained protected. Efforts to mitigate algorithmic bias were prioritized, ensuring fairness and transparency in predictions. Ethical considerations extended to the responsible communication of results, emphasizing the limitations and potential implications of the models.

Thus, the research methodology outlined exemplifies a rigorous approach to understanding and predicting employee attrition. Through the integration of advanced data preprocessing techniques, robust feature engineering, and state-of-the-art machine learning models, the study provides actionable insights for HR professionals. This comprehensive methodology not only enhances the accuracy of attrition prediction but also contributes to the development of proactive workforce management strategies, paving the way for future research in this domain

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Appendix B – Ethics Form



UREC2 RESEARCH ETHICS PROFORMA FOR STUDENTS UNDERTAKING LOW RISK PROJECTS WITH HUMAN PARTICIPANTS

This form is designed to help students and their supervisors to complete an ethical scrutiny of proposed research. The University Research Ethics Policy (<https://www.shu.ac.uk/research/excellence/ethics-and-integrity/policies>) should be consulted before completing the form. The initial questions are there to check that completion of the UREC 2 is appropriate for this study. The final responsibility for ensuring that ethical research practices are followed rests with the supervisor for student research.

Note that students and staff are responsible for making suitable arrangements to ensure compliance with the General Data Protection Act (GDPR). This involves informing participants about the legal basis for the research, including a link to the University research data privacy statement and providing details of who to complain to if participants have issues about how their data was handled or how they were treated (full details in module handbooks). In addition, the act requires data to be kept securely and the identity of participants to be anonymized. They are also responsible for following SHU guidelines about data encryption and research data management. Guidance can be found on the SHU Ethics Website <https://www.shu.ac.uk/research/excellence/ethics-and-integrity>

Please note that it is mandatory for all students to only store data on their allotted networked drive space and not on individual hard drives or memory sticks etc.

The present form also enables the University and College to keep a record confirming that research conducted has been subjected to ethical scrutiny.

The form must be completed by the student and the supervisor and independently reviewed by a second reviewer or module leader (additional guidance can be obtained from your College Research Ethics Chair¹). In all cases, it should be counter-signed and kept as a record showing that ethical scrutiny has occurred. Some courses may require additional scrutiny. Students should retain a copy for inclusion in their research projects, and a copy should be uploaded to the relevant module Blackboard site.

Please note that it may be necessary to conduct a health and safety risk assessment for the proposed research (SECTION B). Further information can be obtained from the [University's Health and Safety Website](#)

SECTION A

¹ College of Social Sciences and Arts - Dr. Antonia Ypsilanti (a.ypsilanti@shu.ac.uk)
College of Business, Technology and Engineering - Dr. Tony Lynn (t.lynn@shu.ac.uk)
College of Health, Wellbeing and Life Sciences - Dr. Nikki Jordan-Mahy (n.jordan-mahy@shu.ac.uk)

1. Checklist questions to ensure that this is the correct form:

Health Related Research within the NHS, or Her Majesty's Prison and Probation Service (HMPPS), or with participants unable to provide informed consent check list.

Question	Yes/No
Does the research involve?	No
• Patients recruited because of their past or present use of the NHS	
• Relatives/carers of patients recruited because of their past or present use of the NHS	No
• Access to data, organs, or other bodily material of past or present NHS patients	No
• Foetal material and IVF involving NHS patients	No
• The recently dead in NHS premises	No
• Prisoners or others within the criminal justice system recruited for health-related research	No
• Police, court officials, prisoners, or others within the criminal justice system	No
• Participants who are unable to provide informed consent due to their incapacity even if the project is not health related	No
• Is this an NHS research project, service evaluation or audit? <i>For NHS definitions please see the following website</i> http://www.hra.nhs.uk/documents/2013/09/defining-research.pdf	No

If you have answered **YES** to any of the above questions, then you **MUST consult with your supervisor** to obtain research ethics from the appropriate institution outside the university. This could be from the NHS or Her Majesty's Prison and Probation Service (HMPPS) under their independent Research Governance schemes. Further information is provided below. <https://www.myresearchproject.org.uk/>

2. Checks for research with human participants

Question	Yes/No
1. Will any of the participants be vulnerable? <i>Note: Vulnerable people include children and young people, people with learning disabilities, people who may be limited by age or sickness, pregnancy, people researched because of a condition they have, etc. See full definition on ethics website in the document Code of Practice for Researchers Working with Vulnerable Populations (under the Supplementary University Policies and Good Research Practice Guidance)</i>	No
2. Are drugs, placebos, or other substances (e.g., food substances, vitamins) to be administered to the study participants or will the study involve invasive, intrusive, or potentially harmful procedures of any kind?	No
3. Will tissue samples (including blood) be obtained from participants?	No
4. Is pain or more than mild discomfort likely to result from the study?	No
5. Will the study involve prolonged or repetitive testing?	No

Question	Yes/No
6. Is there any reasonable and foreseeable risk of physical or emotional harm to any of the participants? <i>Note: Harm may be caused by distressing or intrusive interview questions, uncomfortable procedures involving the participant, invasion of privacy, topics relating to highly personal information, topics relating to illegal activity, or topics that are anxiety provoking, etc.</i>	No
7. Will anyone be taking part without giving their informed consent?	No
8. Is it covert research? <i>Note: 'Covert research' refers to research that is conducted without the knowledge of participants.</i>	No
9. Will the research output allow identification of any individual who has not given their express consent to be identified?	No

If you have answered **YES** to any of these questions you are **REQUIRED** to complete and submit a UREC3 or UREC4 form. Your supervisor will advise. If you have answered **NO** to all these questions, then proceed with this form (UREC2).

3. General project details

Details	
Name of student	Abinaya M
SHU email address	C3085799@hallam.shu.ac.uk
Department/College	Department of Computing
Name of supervisor	Davila Garcia, Maria Luisa
Supervisor's email address	m.l.davila-garcia@shu.ac.uk
Title of proposed research	HR Analytics: Predicting and Addressing Employee Attrition with Machine Learning.
Proposed start date	26/09/2024
Proposed end date	09/01/2025
Background to the study and the rationale (reasons) for undertaking the research (500 words)	Organizations face major issues as a result of employee attrition, which impacts long-term sustainability, cost effectiveness, and productivity. Despite organizational efforts to address it, voluntary turnover—which is frequently caused by low job satisfaction, inadequate compensation, and limited career advancement—remains a chronic problem. A viable option is to use data-driven technologies like machine learning (ML), which allow businesses to examine past employee data and find trends to anticipate and reduce turnover. In order to discover attrition reasons and provide practical retention tactics, this study uses the IBM HR Analytics Employee Attrition and Performance dataset and sophisticated machine learning techniques including Random Forests, Support Vector Machines, and Decision Trees. In addition to improving prediction skills, the research prioritizes

Details	
	ethical issues and puts employee well-being above business results. Organizations may increase employee satisfaction and loyalty by identifying high-risk workers and addressing important issues including tenure, pay, and work-life balance. In order to promote a culture of well-informed decision-making and long-term organizational success, the study also emphasizes the significance of combining data privacy and ethical AI practices, bridging the gap between data science and HR management.
Aims & research question(s)	This project's goal is to use data-driven methods to analyze and forecast staff attrition inside an organization. This project aims to give the human resources department actionable insights by identifying the main causes of attrition. This will allow the department to develop strategies to lower staff turnover and enhance organizational performance. What are the main causes of employee churn in the company, and how can predictive analytics be used to pinpoint vulnerable workers and suggest successful retention tactics?
Methods to be used for: 1. Recruitment of participants 2. Data collection 3. Data analysis	na
Outline the nature of the data held, details of anonymization, storage and disposal procedures as required.	

4. Research in external organizations

Question	Yes/No
1. Will the research involve working with/within an external organization (e.g., school, business, charity, museum, government department, international agency, etc.)?	No
2. If you answered YES to question 1, do you have granted access to conduct the research from the external organization? <i>If YES, students please show evidence to your supervisor. You should retain this evidence safely.</i>	na
3. If you do not have permission for access is this because: A. you have not yet asked B. you have asked and not yet received an answer C. you have asked and been refused access. <i>Note: You will only be able to start the research when you have been granted access.</i>	na

5. Research with products and artefacts

Question	Yes/No
1. Will the research involve working with copyrighted documents, films, broadcasts, photographs, artworks, designs, products, programs, databases, networks, processes, existing datasets, or secure data?	Yes
2. If you answered YES to question 1, are the materials you intend to use in the public domain? <i>Notes: 'In the public domain' does not mean the same thing as 'publicly accessible'.</i> - Information which is 'in the public domain' is no longer protected by copyright (i.e., copyright has either expired or been waived) and can be used without permission. - Information which is 'publicly accessible' (e.g., TV broadcasts, websites, artworks, newspapers) is available for anyone to consult/view. It is still protected by copyright even if there is no copyright notice. In UK law, copyright protection is automatic and does not require a copyright statement, although it is always good practice to provide one. It is necessary to check the terms and conditions of use to find out exactly how the material may be reused etc. <i>If you answered YES to question 1, be aware that you may need to consider other ethics codes. For example, when conducting Internet research, consult the code of the Association of Internet Researchers; for educational research, consult the Code of Ethics of the British Educational Research Association.</i>	The dataset has been taken from Kaggle.
3. If you answered NO to question 2, do you have explicit permission to use these materials as data? <i>If YES, please show evidence to your supervisor.</i>	NA
4. If you answered NO to question 3, is it because: A. you have not yet asked permission B. you have asked and not yet received an answer C. you have asked and been refused access. <i>Note You will only be able to start the research when you have been granted permission to use the specified material.</i>	na

SECTION B

HEALTH AND SAFETY RISK ASSESSMENT FOR THE RESEARCHER

1. Does this research project require a health and safety risk assessment for the procedures to be used? Discuss this with your supervisor and consult the [Risk Assessment Toolkit](#) for teaching research.

- ☐ Yes
☒ No

(If **YES** the completed Health and Safety Risk Assessment form should be attached). You can find a [Blank/Sample Risk Assessment Form](#) at the Checklist, Generic and TORS Risk Assessments on the [Risk Assessment Toolkit](#)

2. Will the data be collected fully online (no face-to-face contact with participants)?

- ☐ Yes (See the safety guidance for online research² and **go to question 8b**).
☒ No (Go to question 3)

3. Will the proposed data collection take place on campus?

- ☐ Yes (Please answer questions 5 to 8)
☒ No (Please complete all questions and consult with your supervisor or HoD for current guidance and permission for face-to-face research outside the university)

4. Where will the data collection take place?

(Tick as many as apply if data collection will take place in multiple venues)

- | Location | Please specify |
|--|-----------------------|
| ☐ Researcher's Residence | |
| ☐ Participant's Residence | |
| ☐ Education Establishment | |
| ☐ Other e.g., business/voluntary organisation, public venue | |
| ☐ Outside UK | |

5. If face-to-face contact with participants is required for your study? Please stipulate below how you will comply with any government requirements related to Covid-19 and social distancing or other limitations on contact.

6. How will you travel to and from the data collection venue?

² Safety guidance for online research includes information on how to set up online surveys and/or conduct online interviews/focus groups. These guidelines can be found in BB. Please check with your supervisor/module leader.

- ☐ On foot ☐ By car ☐ Public Transport
☐ Other (Please specify)

Please outline how you will ensure your personal safety when travelling to and from the data collection venue (include any Covid-19 related precautions)

7. How will you ensure your own personal safety whilst at the research venue?

8. Are there any potential risks to your health and wellbeing associated with either (a) the venue where the research will take place and/or (b) the research topic itself?

- ☒ None that I am aware of
☐ Yes (Please outline below including steps taken to minimise risk)

9. If you are carrying out research off-campus, you must ensure that each time you go out to collect data you ensure that someone you trust knows where you are going (without breaching the confidentiality of your participants), how you are getting there (preferably including your travel route), when you expect to get back, and what to do should you not return at the specified time.

Please outline here the procedure you propose using to do this.

No, my data collection is fully on-line.

Adherence to SHU policy and procedures

Ethics sign-off
Personal statement
I can confirm that: - I have read the Sheffield Hallam University Research Ethics Policy and Procedures - I agree to abide by its principles.
Student

Ethics sign-off	
Name: Abinaya M	Date: 21/11/2024
Signature: <i>Abinaya M</i>	
Supervisor or another person giving ethical sign-off	
I can confirm that completion of this form has not identified the need for ethical approval by the TPREC/CREC or an NHS, Social Care, or other external REC. The research will not commence until any approvals required under Sections 4 & 5 have been received and any necessary health and safety measures are in place.	
Name: Dr Maria Luisa Davila Garcia	Date: 21/11/2024
Signature: 	
Additional Signature if required by course leader:	
Name:	Date:
Signature:	

Please ensure that you have attached all relevant documents. Your supervisor must approve them before you start data collection:

Documents	Yes	No	N/A
Research proposal if prepared previously	☒	☐	☐
Any recruitment materials (e.g., posters, letters, emails, etc.)	☐	☐	☒
Participant information sheet ³	☐	☐	☒
Participant consent form ⁴	☐	☐	☒
Details of measures to be used (e.g., questionnaires, etc.)	☐	☐	☒
Outline interview schedule / focus group schedule	☐	☐	☒
Debriefing materials	☐	☐	☒
Health and Safety Risk Assessment Form	☐	☐	☒

³ It is mandatory to attach the Participant Information Sheet (PIS)

⁴ It is mandatory to attach a Participant Consent Form, unless it is embedded in an online survey, in which case your supervisor must approve it before you start data collection

¹ It is mandatory to attach the Participant Information Sheet (PIS)

¹ It is mandatory to attach a Participant Consent Form, unless it is embedded in an online survey, in which case your supervisor must approve it before you start data collection

Appendix C: GRAPHS, TABLES & CODES

```
✓ 3s # import libraries
import numpy as np
from numpy import median
from scipy import stats
import pandas as pd
import seaborn as sns
sns.set()
import matplotlib.pyplot as plt
%matplotlib inline
from collections import Counter
import random as rn
from functools import reduce

# Import and suppress warnings
import warnings
warnings.filterwarnings('ignore')
import os
```

```
[4] # Read the file
df = pd.read_csv('IBM HR Data new.csv')
df.head()
```

```
5 Unnamed: 0 Age Attrition BusinessTravel DailyRate Department DistanceFromHome Education EducationField EmployeeCount ... RelationshipSatisfaction StandardHours StockOption
0 0 41.0 1 Travel_Rarely 1102.0 Sales 1.0 2.0 Life Sciences 1 ... 1.0 80.0
1 1 41.0 1 Travel_Rarely 1102.0 Sales 1.0 2.0 Life Sciences 1 ... 1.0 80.0
2 2 41.0 1 Travel_Rarely 1102.0 Sales 1.0 2.0 Life Sciences 1 ... 1.0 80.0
3 3 41.0 1 Travel_Rarely 1102.0 Sales 1.0 2.0 Life Sciences 1 ... 1.0 80.0
4 4 41.0 1 Travel_Rarely 1102.0 Sales 1.0 2.0 Life Sciences 1 ... 1.0 80.0
5 rows x 37 columns
```

```
✓ 0s [24] # Check for missing values
df.isnull().sum()
```



	0
Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobRole	0
JobSatisfaction	0
MaritalStatus	0
MonthlyIncome	0
MonthlyRate	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0
RelationshipSatisfaction	0
StandardHours	0
StockOptionLevel	0
TotalWorkingYears	0
TrainingTimesLastYear	0
WorkLifeBalance	0
YearsAtCompany	0
YearsInCurrentRole	0
YearsSinceLastPromotion	0
YearsWithCurrManager	0

```
[25] # Check for duplicate observations
df.duplicated().sum()
```

15

```
duplicate_rows = df[df.duplicated()]
print("\nDuplicate Rows:")
duplicate_rows.head()
```

15

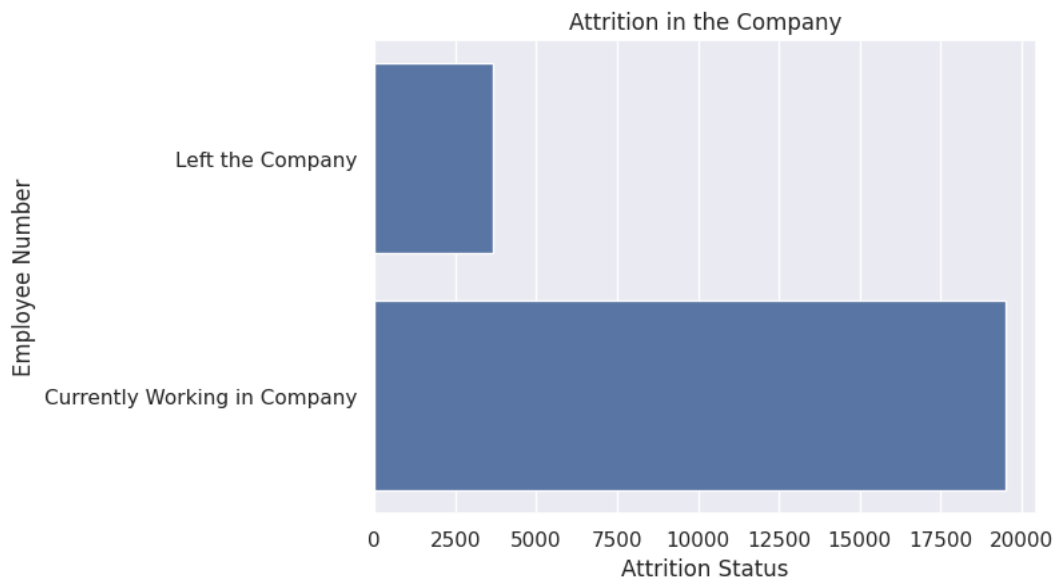
Duplicate Rows:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction
10	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
11	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
12	41	1	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	7	
660	37	1	Travel_Rarely	807	Human Resources	6	4	Human Resources	1	1	
661	37	1	Travel_Rarely	807	Human Resources	6	4	Marketing	1	4	

```
[ ] df2 = df3.drop(labels = ['Over18', 'StandardHours', 'EmployeeCount'], axis = 1)
```

✓
0s

```
[13] # One hot encode categorical dataframe
features_categorical = pd.get_dummies(cat_df, drop_first=True).astype(int)
features_categorical.head()
```



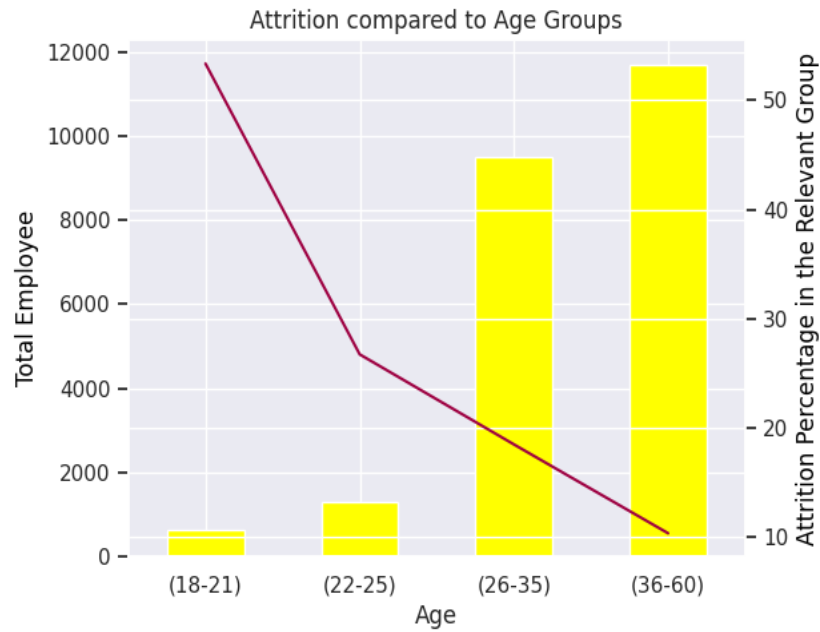
✓
0s

```
[101] # Age Group & Attrition Table
table_df(x = 'Age')
```



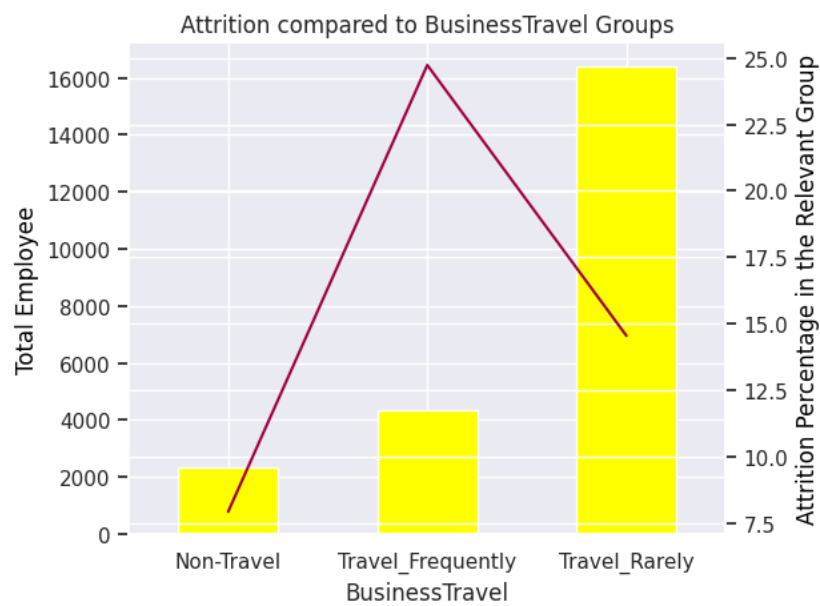
	Age	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	(18-21)	639	341	53.364632	9.334793
1	(22-25)	1306	349	26.722818	9.553791
2	(26-35)	9507	1756	18.470601	48.070079
3	(36-60)	11687	1207	10.327715	33.041336





```
[103] # Business Travel & Attrition Table
table_df(x = 'BusinessTravel')
```

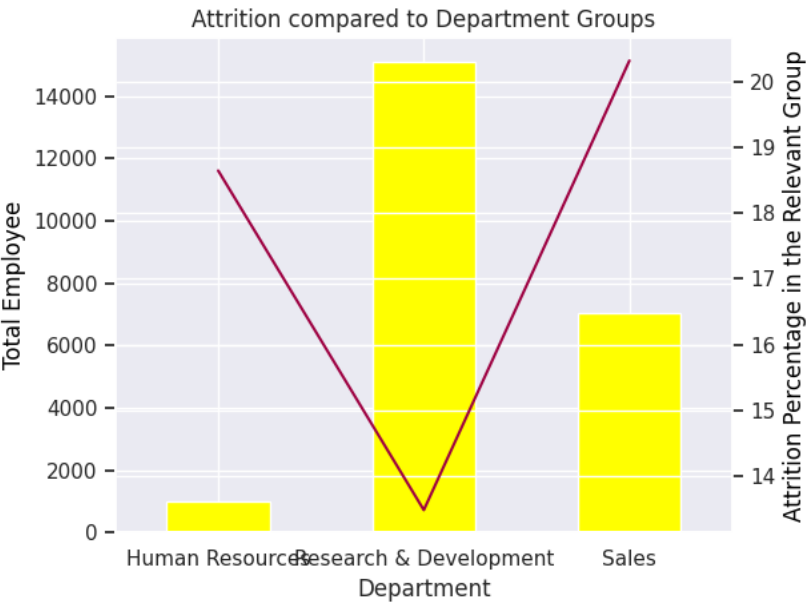
	BusinessTravel	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Non-Travel	2362	187	7.917019	5.119080
1	Travel_Frequently	4367	1080	24.730937	29.564741
2	Travel_Rarely	16410	2386	14.539915	65.316178



✓
0s

[105] # Department & Attrition Table
table_df(x = 'Department')

	Department	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Human Resources	1003	187	18.644068	5.119080
1	Research & Development	15097	2036	13.486123	55.735012
2	Sales	7039	1430	20.315386	39.145907

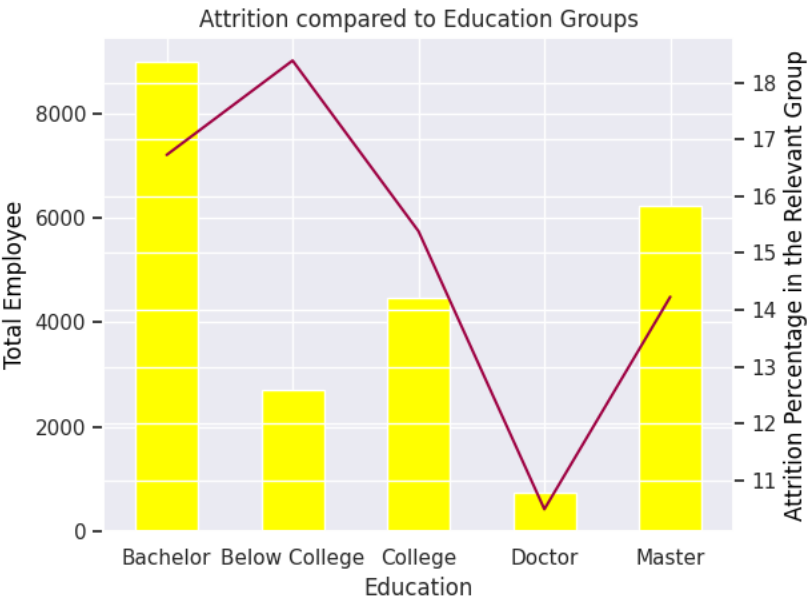


Education

✓
0s

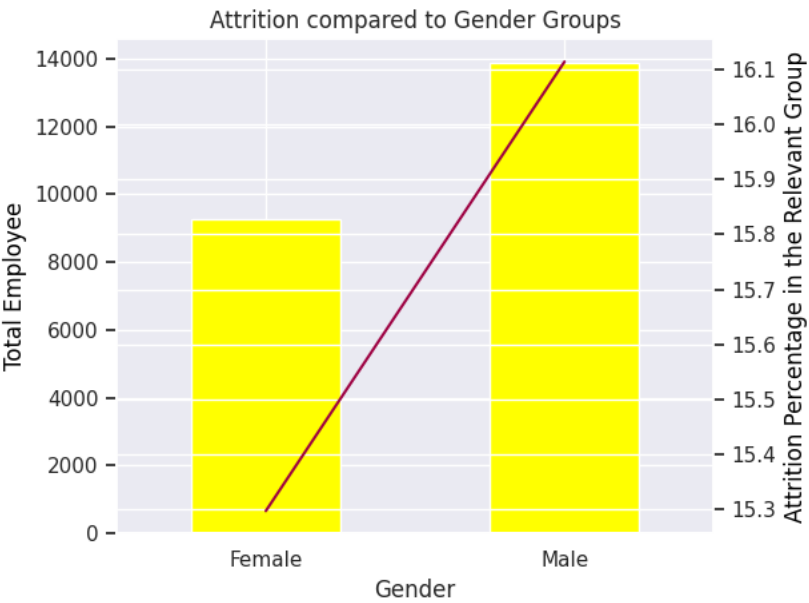
[111] # Education Level & Attrition Table
table_df(x = 'Education')

	Education	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Bachelor	8984	1503	16.729742	41.144265
1	Below College	2697	496	18.390805	13.577881
2	College	4465	687	15.386338	18.806460
3	Doctor	753	79	10.491368	2.162606
4	Master	6240	888	14.230769	24.308787



```
[118] # Gender & Attrition Table
table_df(x = 'Gender')
```

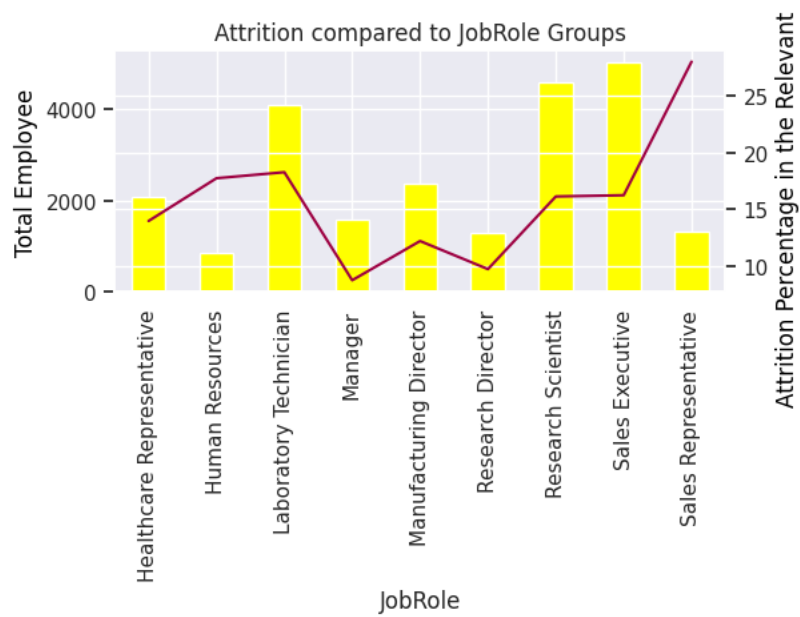
	Gender	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Female	9257	1416	15.296532	38.762661
1	Male	13882	2237	16.114393	61.237339



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```
[126] # Job Role & Attrition Table
table_df(x = 'JobRole')
```

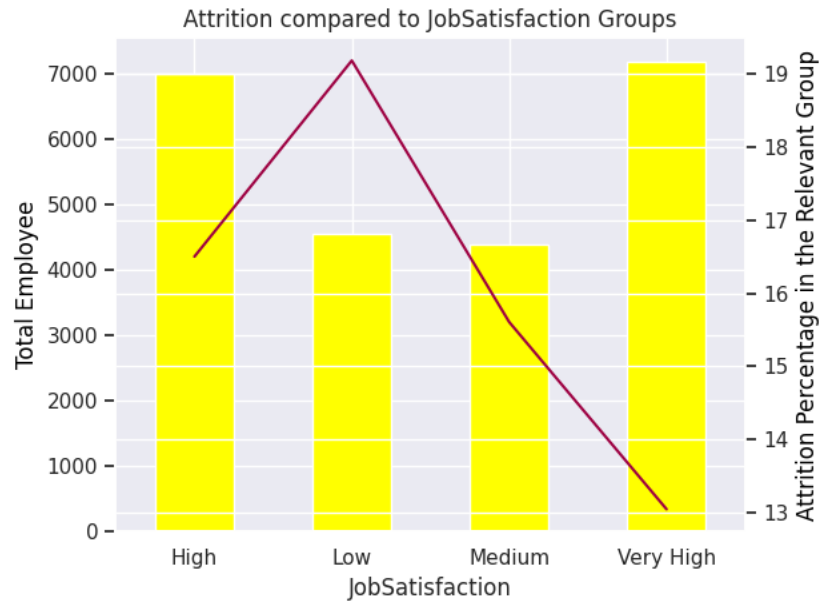
	JobRole	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	Healthcare Representative	2072	289	13.947876	7.911306
1	Human Resources	835	148	17.724551	4.051465
2	Laboratory Technician	4097	748	18.257261	20.476321
3	Manager	1584	138	8.712121	3.777717
4	Manufacturing Director	2357	287	12.176496	7.856556
5	Research Director	1279	124	9.695074	3.394470
6	Research Scientist	4580	738	16.113537	20.202573
7	Sales Executive	5036	817	16.223193	22.365179
8	Sales Representative	1299	364	28.021555	9.964413



```
[128] # Labeling Job Role
table_df(x = 'JobSatisfaction')
df2['JobSatisfaction'] = df2['JobSatisfaction'].map({1:'Low', 2:'Medium',
3:'High', 4:'Very High'})
```

```
[129] # Job Satisfaction & Attrition Table
table_df(x = 'JobSatisfaction')
```

	JobSatisfaction	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	High	7005	1156	16.502498	31.645223
1	Low	4551	873	19.182597	23.898166
2	Medium	4395	686	15.608646	18.779086
3	Very High	7188	938	13.049527	25.677525

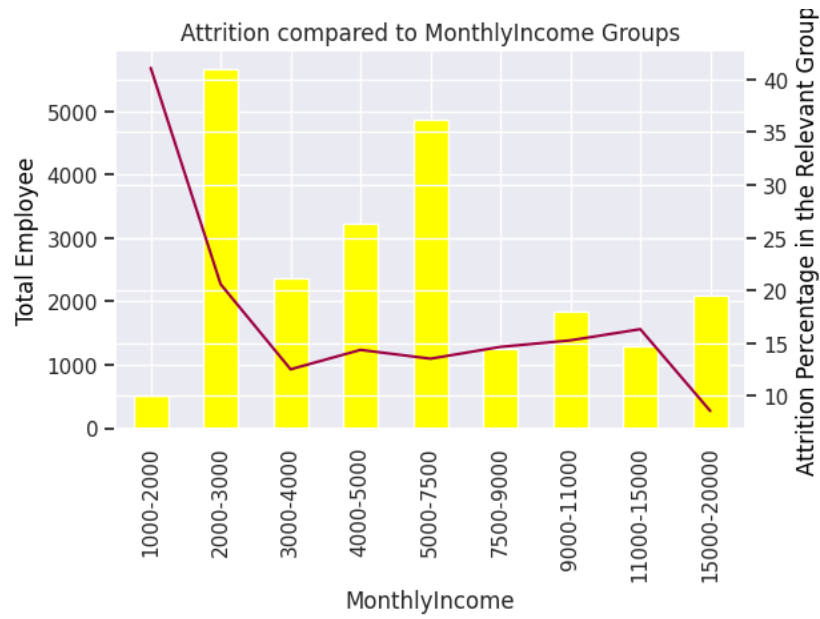


```
[133] # Labeling Monthly Income Level
table_df(x = "MonthlyIncome")
bins = [1000, 2000, 3000, 4000, 5000, 7500, 9000, 11000, 15000, 20000]
group_names = ['1000-2000', '2000-3000', '3000-4000', '4000-5000', '5000-7500',
               '7500-9000', '9000-11000', '11000-15000', '15000-20000']
df2["MonthlyIncome"] = pd.cut(df2["MonthlyIncome"], bins, labels = group_names)
```

```
[134] # Monthly Income & Attrition Table
table_df(x = "MonthlyIncome")
```

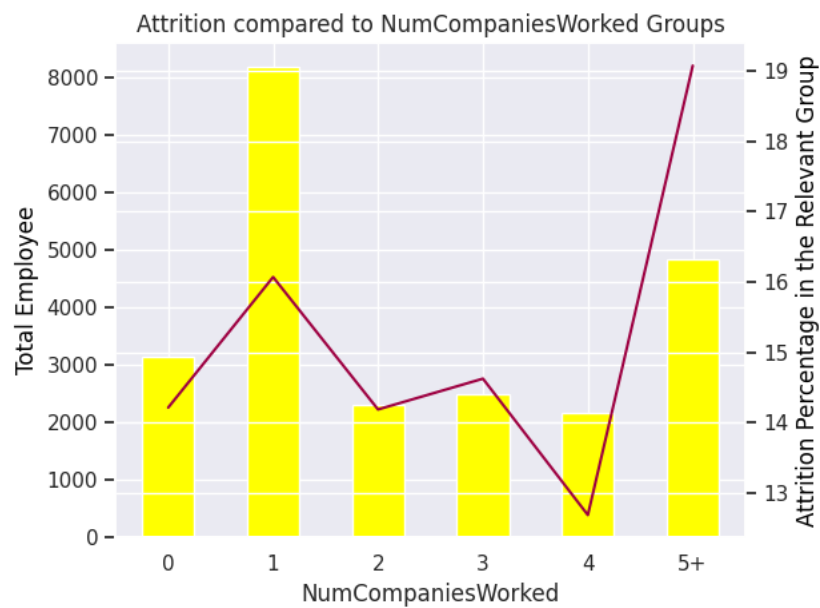
	MonthlyIncome	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	1000-2000	516	212	41.085271	5.803449
1	2000-3000	5675	1168	20.581498	31.973720
2	3000-4000	2357	295	12.515910	8.075554
3	4000-5000	3225	463	14.356589	12.674514
4	5000-7500	4874	659	13.520722	18.039967
5	7500-9000	1250	183	14.640000	5.009581
6	9000-11000	1848	282	15.259740	7.719682
7	11000-15000	1293	211	16.318639	5.776074
8	15000-20000	2101	180	8.567349	4.927457

Activate Windows



```
[137] # Numbers Companied Worked & Attrition Table
table_df(x = 'NumCompaniesWorked')
```

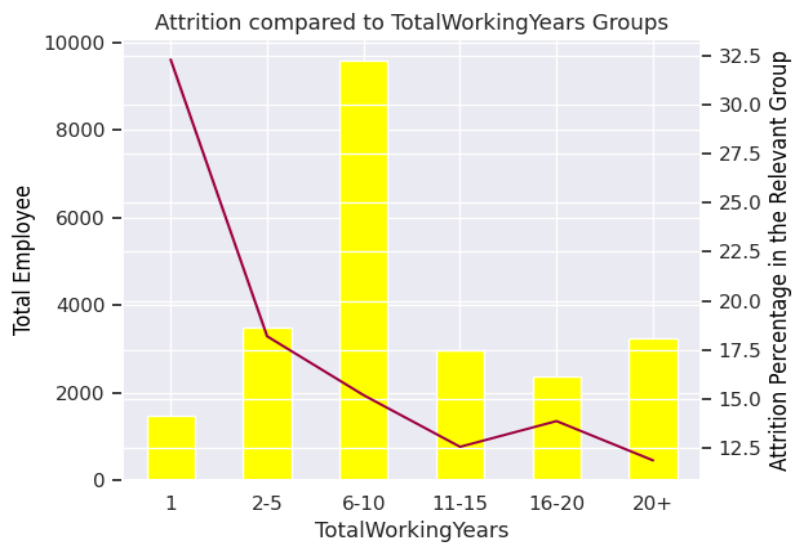
	NumCompaniesWorked	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	0	3138	446	14.212874	12.209143
1	1	8195	1317	16.070775	36.052560
2	2	2298	326	14.186249	8.924172
3	3	2489	364	14.624347	9.964413
4	4	2176	276	12.683824	7.555434
5	5+	4843	924	19.079083	25.294279



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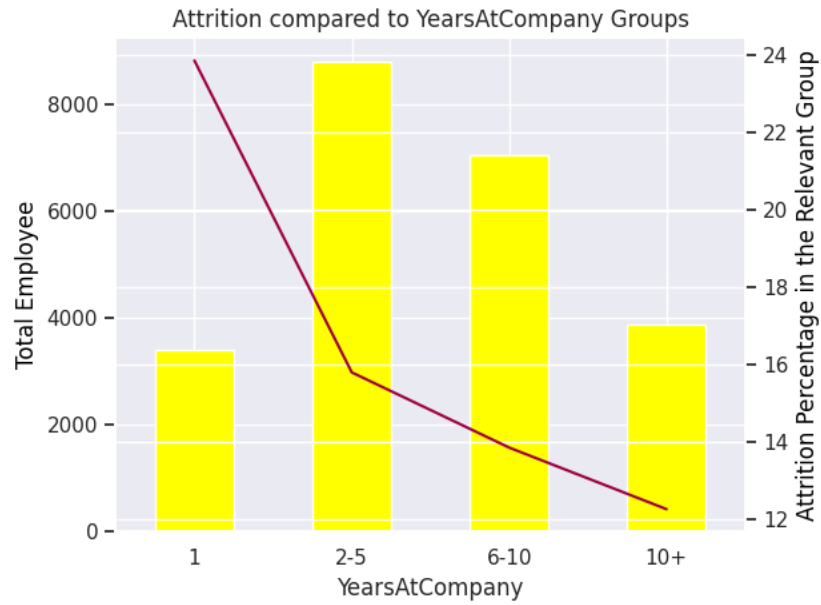
```
[152] # Total Working Years & Attrition Table
table_df(x = 'TotalWorkingYears')
```

	TotalWorkingYears	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	1	1465	473	32.286689	12.948262
1	2-5	3502	637	18.189606	17.437722
2	6-10	9585	1456	15.190402	39.857651
3	11-15	2978	374	12.558764	10.238160
4	16-20	2365	328	13.868922	8.978921
5	20+	3244	385	11.868064	10.539283



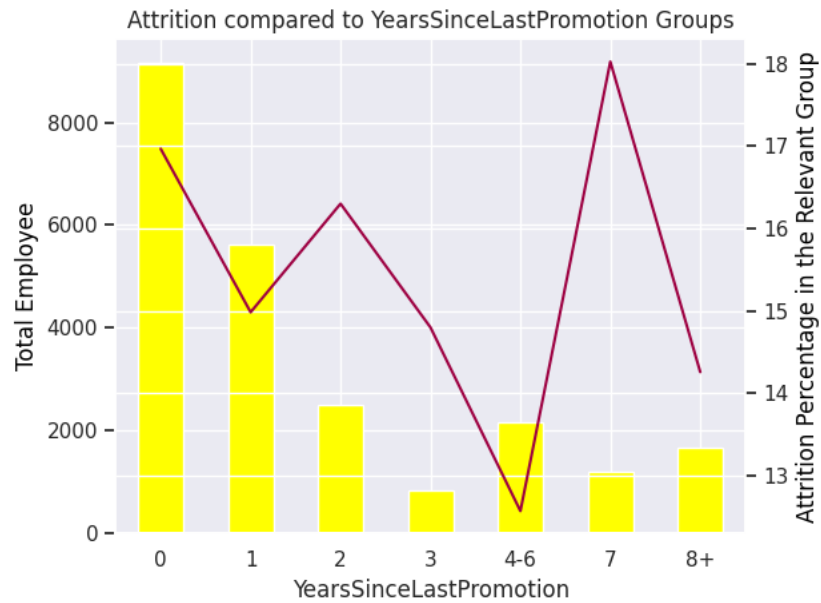
```
[160] # Years at Company & Attrition Table
table_df(x = 'YearsAtCompany')
```

	YearsAtCompany	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	1	3408	813	23.855634	22.255680
1	2-5	8799	1389	15.785885	38.023542
2	6-10	7045	975	13.839603	26.690391
3	10+	3887	476	12.245948	13.030386

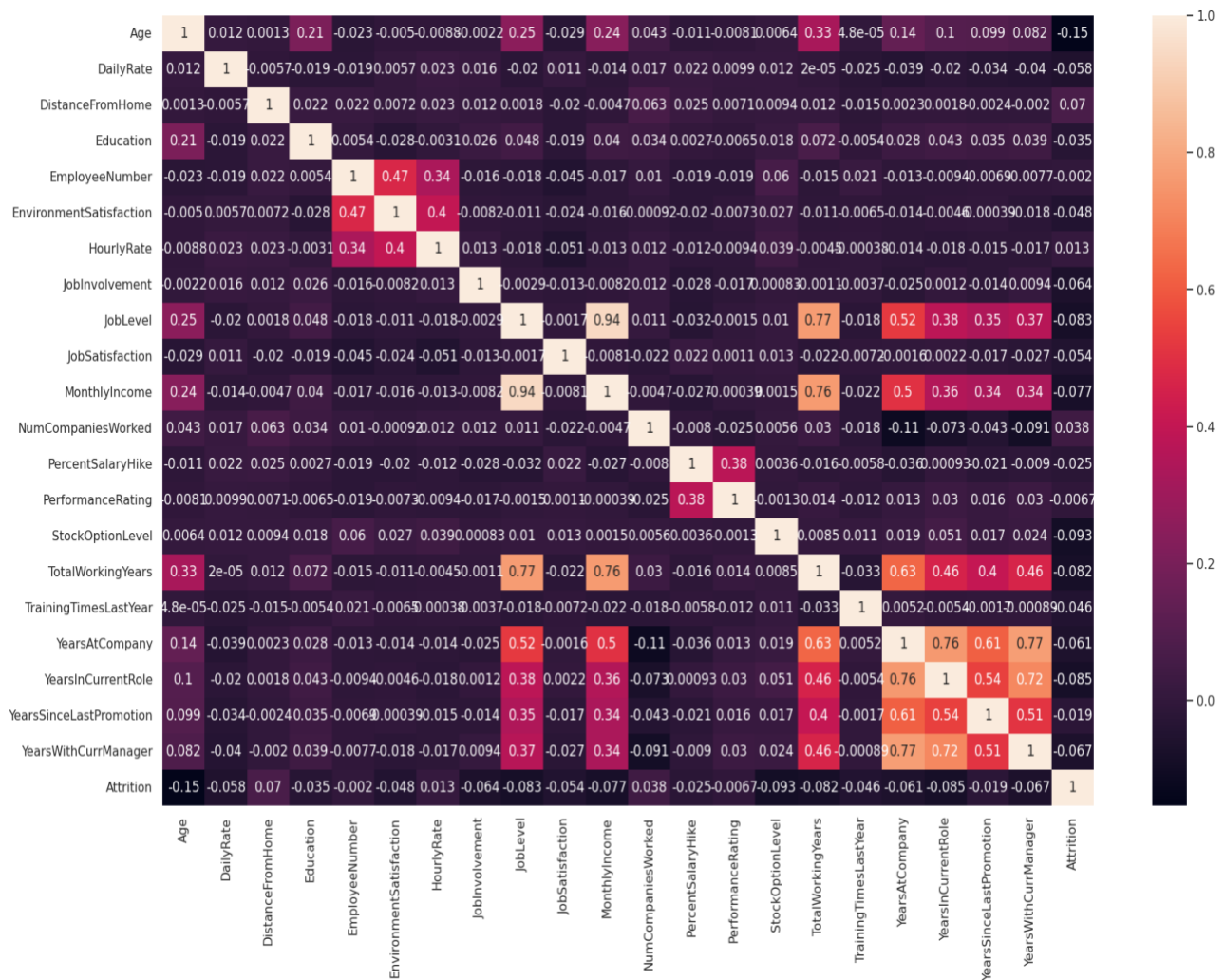


```
[166] # Years Since Last Promotion & Attrition Table
table_df(x = 'YearsSinceLastPromotion')
```

	YearsSinceLastPromotion	Total Employee	Attrition Number	% of Attrition in the Relevant Group	% of Attrition in the Company
0	0	9159	1554	16.966918	42.540378
1	1	5612	841	14.985745	23.022174
2	2	2509	409	16.301315	11.196277
3	3	831	123	14.801444	3.367096
4	4-6	2155	271	12.575406	7.418560
5	7	1204	217	18.023256	5.940323
6	8+	1669	238	14.260036	6.515193



HR Analytics: Predicting and Addressing Employee Attrition with Machine Learning



Feature	Description	Type
Age	Numerical representation of the employee's age, capturing career stage and retirement proximity.	Numeric
Attrition	Target variable indicating whether the employee left the organization (Yes = 1, No = 0).	Categorical
BusinessTravel	Frequency of business travel categorized as No Travel, Travel Rarely, and Travel Frequently.	Categorical
DailyRate	Represents the daily salary rate of the employee, highlighting compensation variations.	Numeric
Department	Department where the employee works, such as HR, R&D, or Sales, indicating functional alignment.	Categorical

DistanceFromHome	The distance (in kilometers) between the employee's residence and the workplace, often correlating with satisfaction levels.	Numeric
Education	Education level on a scale (e.g., High School = 1, Bachelor = 2), reflecting academic qualifications.	Numeric
EducationField	Employee's field of study, including Life Sciences, Marketing, and Technical Degree, indicating specialization.	Categorical
EnvironmentSatisfaction	Satisfaction with the work environment, rated on a scale of 1 to 4, affecting engagement and productivity.	Numeric
Gender	Gender of the employee (Male/Female).	Categorical
HourlyRate	Hourly wage of the employee, useful for assessing compensation fairness.	Numeric
JobInvolvement	Extent of the employee's involvement in their role, rated from 1 (low) to 4 (high).	Numeric
JobLevel	Hierarchical level of the job (e.g., entry-level, mid-level), indicating seniority and career progression.	Numeric
JobRole	Specific job designation, such as Manager, Sales Executive, or Research Scientist, reflecting role diversity.	Categorical
JobSatisfaction	Employee's satisfaction with their job, rated on a scale of 1 to 4.	Numeric
MaritalStatus	Marital status categorized as Single, Married, or Divorced, potentially influencing work-life dynamics.	Categorical
MonthlyIncome	Employee's total monthly salary, highlighting compensation.	Numeric
OverTime	Indicates whether the employee works overtime (Yes/No), often correlating with job stress and attrition.	Categorical

PerformanceRating	Performance appraisal score (1 to 4), indicating efficiency and contribution.	Numeric
RelationshipSatisfaction	Satisfaction with interpersonal relationships at work, rated from 1 to 4.	Numeric
StockOptionLevel	Stock options provided, reflecting organizational incentives for retention.	Numeric
TotalWorkingYears	Total professional experience in years, indicating career maturity.	Numeric
TrainingTimesLastYear	Number of training sessions attended, indicating opportunities for skill development.	Numeric
WorkLifeBalance	Perceived balance between work responsibilities and personal life, rated from 1 to 4.	Numeric
YearsAtCompany	Number of years spent in the current organization, indicating organizational loyalty.	Numeric
YearsInCurrentRole	Tenure in the current role, reflecting job stability and growth.	Numeric
YearsSinceLastPromotion	Years since the employee's last promotion, highlighting career advancement.	Numeric
YearsWithCurrManager	Number of years under the current manager, reflecting team dynamics and satisfaction.	Numeric